**SmartLink: A Unified Platform for IT Services and Customer Support**

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In Partial Fulfillment of the

Requirements for the Degree of

Bachelor of Science in Information Technology

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To SMFP (Services for Mobile and PC Fix), our deepest appreciation for allowing us to collaborate and use their organization as a case study for our research. Thank you for your cooperation, insights, and for trusting our proposed system.

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## ABSTRACT

SmartLink is a new project that develops and integrates the Tickets and Repair Management System with the Business System for SMFP, a local computer reparative and services-oriented business. This is one of the ways the company is able to solve problems related to responding to customers' inquiries, managing job orders, and improving the company's online presence.

Currently, SMFP relies heavily on Facebook for communicating with clients, a limitation that increases the ineffectiveness of customer service and repair track. Customers are dissatisfied due to frequent missed deadlines, poor tracking of documents, and continued absence of a structured ticketing workflow.

Modules covered by the proposed system are Customer Inquiry Portal, Automated Ticketing, Repair Job Order Tracker, Technician Assignment Panel. Along with that, it has a website for allowing customers to book online, estimate costs, and troubleshoot issues interactively. The admin dashboard contains analytics, automated updates, and repair histories to enable better management.

This project also adopted the Software Development Life Cycle (SDLC) approach for its development and incorporated data flow diagrams, ERD, use case diagrams, and system flowcharts in support of its design. Preliminary evaluations indicate the improvement of customer engagement, response time reduction, and a more professional digital presence of the company

**TABLE OF CONTENTS**


| | |
|---|---|
| APPROVAL SHEET | ii |
| ACKNOWLEDGEMENT | iii |
| ABSTRACT | iv |
| TABLE OF CONTENTS | v |
| LIST OF FIGURES | vi |
| LIST OF TABLES | vii |
| **Chapter I. Introduction** | |
| Project Purpose | 2 |
| Project Description | 2 |
| Objective of the Study | 3 |
| General Objective of the Study | 3 |
| Specific Objectives of the Study | 3 |
| Significance of the Study | 4 |
| Scope and Delimitation of the Study | 4 |
| Scope | 4 |
| Limitation | 5 |
| Review of Related Literature | 5 |
| Review of Related Studies | 6 |
| Technical Background | 8 |
| Conceptual Framework | 8 |
| Definition of Terms | 11 |
| **Chapter II. Methodology** | |

Design of Software, System, Product or Process 16
Software Architecture 17
Data Flow Diagram 20
Software Development Life Cycle 31
System Flowchart 33
Entity Relationship Diagram 45
Requirements Analysis 49
Requirements Documentation 51
Development and Pre-Testing procedures 55
Implementation Plan (Infrastructure/ Deployment) 57
Statistical Treatment of data 60
Data Gathering Technique 60
Setting of the Study 61
Subject of the Study 61
Sampling Technique 61
**Chapter III. Results and Discussion**
System Test Plan 63
Statistical Analysis 63
Implementation and Statistical Results Interpretation 71
**Chapter IV. Summary of Findings, Conclusion, and Recommendation**
Summary 93
Recommendations 94
Conclusions 95

**Bibliography (References)**

**Appendices**

Appendix A: Letter Request for Respondents

Appendix B: Evaluation Instrument (User and IT Expert)

Appendix C: Summary of Evaluation Result

Appendix D: Test Result

Appendix E: System Screenshot

Appendix F: User Manual

Appendix G: Similarity Index (Turnitin)

Appendix H: Grammarian (Grammarian Certificate)

Appendix I: Curriculum Vitae (One Page per Student)

**LIST OF FIGURES**

Figure No.1: IPO with Evaluation Model Conceptual Framework

Figure No.2: Software architecture

Figure No.3:  Data Flow Diagram - Level 0 (Yourdon & DeMarco, 1980)

Figure No.4: Data Flow Diagram - Level 1 (Yourdon & DeMarco, 1980)

Figure No.5: Data Flow Diagram - Level 2 (Yourdon & DeMarco, 1980)

Figure No.6: Software Development Cycle

Figure No.7: Customer System Flowchart

Figure No.8: Payment System Flowchart

Figure No.9: Admin System Flowchart

Figure No.10: Technician System Flowchart

Figure No.11: SmartLink's Entity Relationship Diagram

**LIST OF TABLES**

Data Flow Table……………………………………………..…….……………25

Customer Table…………………..………………………..…………………47

Inquiry Table……………………..…………………………..…………………48

Booking Table…………………..………..………………..…………………48

Repair Job Table…………………..……………………………………………49

Likert Scale Table…………………..……………….……………..……………62

Functional Effectiveness Testing Procedures…..…………………………………66

Usability Testing Procedures……………………………..………..…..…………68

User Respondents…………………..…………………………..……..………76

II Expert Respondents…………………..……………………..…….…..………82

# Chapter I

## Introduction

In a digital-centric business landscape, organizations are pressured to enhance communication and customer service for sustained growth. This project aims to create a Ticketing and Repair Management System coupled with an Integrated Business Website for SMFP, a repair and IT services firm. The objective is to improve customer interaction, service tracking, and business visibility. Currently, SMFP utilizes Facebook for service bookings, which, despite its wide reach, lacks the necessary structure for organizing repair requests and delivering timely updates. Consequently, this results in delays and an inadequate level of personalized service, ultimately leading to customer attrition.

By shifting from a manual, social media-based approach to a comprehensive digital platform, SMFP can significantly enhance operational efficiency and customer satisfaction. The proposed system aims to streamline workflows, data management, and foster customer trust through transparent and reliable communication. In today's competitive service industry, such a user-centric digital infrastructure is crucial for sustainable growth and adaptability.

This Capstone Project outlines a solution that allows users to log technical issues, book repairs, track service requests, and access company services via an interactive website. Key features of the platform will include a Ticketing System, Repair Tracking, Customer Notifications, an Admin Dashboard, and essential website elements like an online booking form.

**Project Purpose**

The purpose of this project is to develop a web-based Ticketing and Repair Management System with an Integrated Business Website for SMFP Computer to address the operational inefficiencies caused by manual and social media–based customer service processes. The current method of handling inquiries and repair bookings through Facebook results in delayed responses, disorganized repair records, inconsistent customer communication, and difficulties in monitoring repair progress. This project aims to provide a centralized digital platform that will automate ticket management, repair tracking, technician assignment, customer notifications, quotation generation, and service monitoring. The system is intended to improve operational efficiency, enhance customer satisfaction, strengthen SMFP Computer's online presence, and support more organized and reliable IT service management.

This project aims to improve operational efficiency, customer engagement, and overall service quality through digital transformation.

**Project Description**

This project involves the design and development of a web-based Ticketing and Repair Management System integrated with a business website for SMFP Computer. The system will serve as a centralized platform where customers can submit repair requests, book services, track repair progress, receive notifications, and access company information online.

The system will include modules such as customer management, ticket generation, repair tracking, technician assignment, quotation generation, payment management, reporting, and administrative monitoring. Customers will be able to monitor the status of their repair requests in real time, while administrators and technicians will have tools for organizing repair jobs, managing customer records, and generating service reports efficiently.

The proposed system will be developed to improve workflow automation, reduce manual processing, enhance communication between customers and technicians, and provide a professional digital presence for SMFP Computer. The platform will be accessible through desktop and mobile web browsers and will support Cash and GCash payment methods with manual verification for GCash transactions.

**Objectives of the Study**

The General Objective of this study is to develop an integrated ticketing and repair management system, accompanied by a dynamic corporate Web site, to ease the customer service operations, automate the tracking of repairs, and form an impressive digital front for SMFP.

**Specific Objectives of the study**

- To establish a new Ticketing System to log and classify issues and enable customers to submit repair issues simply through form or interface. Categorize tickets by type and urgency
- To automatically get assigned the tickets to the available technician, based on current workload or specialization. It is meant to automate notification and updates with real-time information.
- To notify customers through email for ticket creation and booking confirmation. Repair Tracking Module is to be implemented inclusive of the technician and job order management. This module should include tracking

for repaired jobs both ongoing and completed. Monitoring of technician performance and completion timelines of jobs.

- To centralize data management and provide searchable records. Store customer information, repair histories, and tickets in a well-secured database. Make conversational repair records for customers and administrators. Ready-to-use automated reports for analysis and decision-making.
- To implement a module for generating service quotations based on the customer's selected service type, required parts, and labor rates—automating a previously manual process. Integrate a secure payment system that supports popular methods such as GCash and Cash to ensure convenient and transparent transactions.

**Significance of the Study**

This study is significant as it addresses critical operational inefficiencies within SMFP Computer. By integrating a modern support system and professional website, it improves customer satisfaction, employee workflow, and the company's reputation. The proposed system provides real-time tracking, structured communication, and user-friendly tools that enhance both internal management and customer interaction.

**SMFP Computer's Management Team -** The management team will benefit from improved customer reach, streamlined communication, and enhanced service visibility, helping them attract more clients and manage inquiries efficiently.

**SMFP Computer's Customer Service Staff -** Staff will experience improved issue tracking, faster resolution times, and an organized workflow for handling customer concerns.

**Customers Seeking IT Solutions, Repairs, or Computer Accessories –** Customers will benefit from easier access to information, efficient service bookings, and timely updates on repair progress.

**Technicians and Repair Staff -** Technicians will benefit from organized repair tracking, clear job assignments, and a searchable repair history database, improving workflow.

**Local Businesses and Partners -** Improved web presence and streamlined customer support may enhance partnerships with local businesses seeking IT solutions

**Future Researcher** - This study provides a foundation for future researchers to develop IT service platforms, examining methodologies, architecture, and user interface design in SmartLink. It encourages exploration into scalable, user-centric platforms, incorporating emerging technologies, and improving interoperability.

**Scope and Delimitation of the Study**

**Scope**

- A business website with a contact form, and booking system. A ticketing system, automated prioritization, and reporting. Also, repair job management, predictive maintenance, and digital report.

**Limitation**

- While the *SMFP SmartLink* system is designed to enhance IT service and customer support workflows, certain limitations were intentionally defined to keep the scope manageable and aligned with the project's core objectives:

- No E-Commerce Functionality The system does not include a full-featured e-commerce platform. Customers will not be able to purchase items directly (such as repair parts or service packages) through the system. All transactions and purchases are expected to occur through separate offline or external channels.
- Absence of Advanced CRM Features Although the system supports customer ticketing and communication, it does not integrate advanced Customer Relationship Management (CRM) tools such as marketing automation, customer profiling, or segmentation analytics. These features fall outside the intended focus on service management and technical support.
- No Native Mobile Application The system is built as a web-based platform, optimized for mobile browsers. However, there is no dedicated mobile app for iOS or Android at this stage. Users can still access the platform via their mobile web browsers due to its responsive and mobile-friendly design.
- The system supports only Cash and GCash payment methods; other digital payment options such as credit cards, online banking, or fully automated payment gateways are not included, and GCash transactions require manual verification by technicians.
- The system provides basic reporting and analysis and does not include AI- or machine-learning–based predictive maintenance features.
- The system implements basic security and access controls and does not support enterprise-level security or large-scale scalability.

These limitations allow the development team to concentrate on delivering a robust, streamlined platform for IT support and service management while leaving room for potential future expansion**.**

## Review of Related Literature

### Ticketing Systems in IT Service Management

Fuchs, Drieschner and Wittges (2022). In their literature review, researchers highlighted the advancements in automated support ticket systems, emphasizing the integration of machine learning to enhance efficiency and accuracy in handling customer inquiries.

Garg and Deshmukh (2020) conducted a comprehensive review of maintenance management literature, identifying key issues ranging from optimization models to information systems, which are crucial for effective maintenance strategies.

Aglibar and Alegre (2022). This study examined the utilization of the JIRA ticketing system in IT companies, concluding its effectiveness in tracking incidents and requests, despite some limitations, and recommending exploration of other analytic models for enhanced results.

Lakshanya and Vaidehi (2024). The article discusses how ticketing systems serve as integral IT service management platforms, facilitating remote assistance and streamlining service requests to ensure efficiency across various organizations.

### Benefits of Help Desk Ticketing Systems

HappyFox (2024). This resource outlines the advantages of help desk ticketing systems, including improved customer satisfaction through timely responses, status updates, and self-service options, thereby enhancing overall support efficiency.

Journal of Theoretical and Applied Information Technology. (2024). This study explores various machine learning algorithms to improve help desk ticketing systems, focusing on classification, clustering, and regression techniques to enhance ticket categorization and prioritization. Machine learning approaches for helpdesk ticketing systems.

National Academies of Sciences, Engineering, and Medicine. (2020). The National Academies of Sciences, Engineering, and Medicine discuss the implementation of electronic ticketing systems in construction management, highlighting their benefits in efficiency and accuracy. Electronic ticketing of materials for construction management.

Regis University. (2023). This thesis examines the development and implementation of a ticketing system tailored for small businesses, addressing specific challenges and proposing solutions to improve customer service.

**Predictive Maintenance Systems**

Zhu, T., Ran, Y., Zhou, X., & Wen, Y. (2019). A survey of predictive maintenance: Systems, purposes and approaches. A comprehensive survey on predictive maintenance systems, discussing their architectures, optimization objectives, and methods, emphasizing the role of machine learning in enhancing maintenance strategies.

ScienceDirect. (2022) This paper provides a bibliometric analysis and literature review of maintenance policies and models, focusing on reuse and remanufacturing strategies to improve sustainability in maintenance practices.

## Review of Related Studies

### Computerized Maintenance Management Support

Kans, M. (2020). This literature review investigates the advancement of computerized maintenance management systems (CMMS), analyzing their development over four decades and their impact on maintenance practices. A literature review of computerized maintenance management support

Simões, J., Gomes, C., & Yasin, M. (2021). This study reviews various methods for measuring maintenance performance, discussing their applications and effectiveness in different organizational contexts. A literature review of maintenance performance measurement.

### Management of Repair and Maintenance Work

ESTS Journal. (2023). This article examines how supervisors manage repair and maintenance work in large organizations, emphasizing the importance of organizational knowledge and resource management. The management of repair and maintenance work in large organizations.

Miller, K., & Dubrawski, A. (2020). This paper reviews current literature on system-level predictive maintenance, identifying gaps and proposing a holistic modeling approach to improve maintenance strategies for complex assets.

Ogunfowora, O., & Najjaran, H. (2023). This study explores the application of reinforcement and deep reinforcement learning in maintenance planning, discussing how these approaches can optimize maintenance schedules and policies. Reinforcement and deep reinforcement learning-based solutions for machine maintenance planning, scheduling policies, and optimization.

Philippine EJournals. (2020). This study focused on creating a system that allows users to generate repair service reports and track inventory stocks, aiming to streamline the repair process and improve customer satisfaction in the automotive sector. Developing a repair management system with inventory management.

**Maintenance Management System**

Mulyono, D., & Sutrisno, E. (2023). The article analyzes root errors in maintenance management and emphasizes the application of selected philosophies to enhance maintenance processes, advocating for a systemic approach to achieve desired performance levels. Maintenance management system and root cause analysis.

Mobley, R. K. (2020). This paper discusses the role of Computerized Maintenance Management Systems (CMMS) in facilitating increased performance and dependability, highlighting their efficiency in process management.

Sorooshian, S. (2023). The study conducts a comprehensive literature review to understand how Industry 4.0 technologies affect various functions of maintenance management systems, aiming to integrate these advancements for improved operations. Integration of maintenance management system functions using Industry 4.0 technologies

Ferreira, C. B., & Moreira, M. F. (2022). This research highlights the necessity of maintenance management systems in addressing evaluation processes' needs, based on bibliometric and interviews with professionals in Brazilian public institution.

## Conceptual Framework

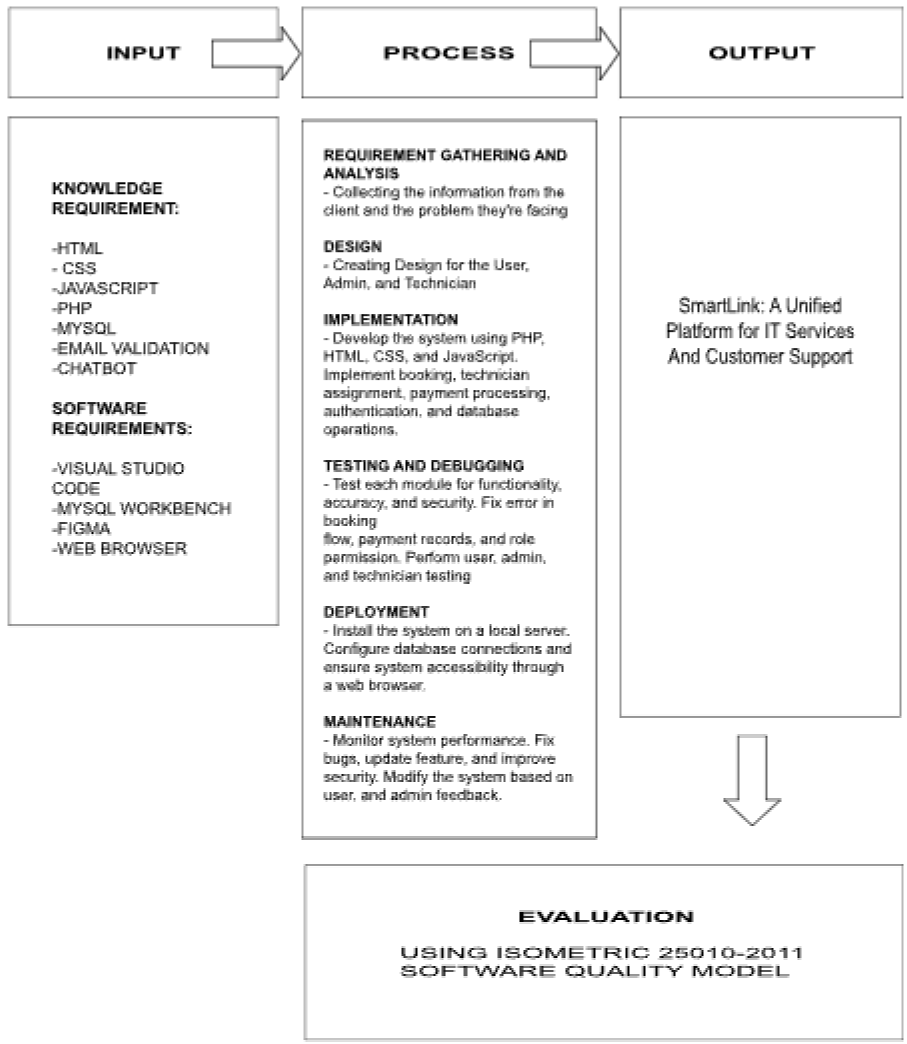



***Figure No. 1**: Input, Process, and Output of the Proposed System*

## Input

The input phase involves all equipment, knowledge, software, and hardware requirements for the building of the system. Knowledge-wise, system development implies knowledge of front-end and back-end technologies, namely HTML, CSS, Tailwind CSS, JavaScript, PHP, MySQL, GIT for version control. It also requires knowledge of repair and ticketing workflows, RESTful APIs, and software quality standards under ISO 25010.

The developers' software tools will include Visual Studio Code for writing and editing code, MySQL Server for managing databases, and Postman for API testing. Figma will be used for designing the user interface and experience of the system. A system will be deployed and tested on computer environments running either Windows or macOS while optionally using a smartphone to test mobile responsiveness

The hardware requirement includes a computer or laptop with a minimum of an Intel i3 8th Gen processor and 8GB of RAM, and a smartphone (Android or iOS) for responsive testing

**Process**

In the Planning phase, the goals of the project are established: to automate responses from the customer support division, to track repair job orders, and to design and implement a web-based system for managing tickets and service requests.

The System Analysis phase aims to collect data regarding the current manual semi-automated repair process at SMFP. Certain pain points will be noted, including delayed responses, missing records, and lack of a centralized repair tracking method. Requirements will be gathered through interviews and surveys conducted with stakeholders.

During the System Design phase, the architecture of the system will be laid out. This includes the preparation of data-flow diagrams (DFDs), entity-relationship diagrams (ERDs), and user interface designs for each module: Inquiry Module, Repair Tracking Module, Status Notification Module, Admin Dashboard, and Customer Portal.

The Development stage refers to coding the system with the help of PHP, MySQL, and JavaScript. Key highlights include online ticketing, and automated notifications.

The Testing phase involves different levels of quality assurance: functional testing (to check if all features function properly), usability testing (to ensure good experience for users), compatibility testing (to support interworking on different browsers and devices), and acceptance testing (to test whether the system has met expectations of stakeholders).

Finally, during Deployment and Maintenance, the system will be deployed onto a server or hosting platform and monitored for performance. Thereafter, the Maintenance phase consists of bug fixes, updates, and any form of support required by user post-launch.

**Output**

The output of the operation is the functional end-to-end web-based ticketing and repair management solution Fixerman System that optimizes service operations, fastens communication between customer businesses and technicians, and automates repair-related workflows. The system comes with a support portal, online booking system, real-time status updates, repair cost estimators, and admin dashboards with service analytics. It streamlines the operations of repair service while delivering an exceptional user experience.

Testing is carried out using an evaluation model standard ISO 25010 Software Quality Model: Based on its functionality, reliability, usability, efficiency, maintainability, and portability. This ensures that the SmarLink System is at part with high standards of software quality and performance levels especially in letting users easily and efficiently handle all the issues related to repairs.

**Definition of Terms**

**Analytics Module** An administrative tool that provides insights into service trends, customer behavior, and peak booking periods to support data-driven decision-making.

**Customer Portal** A user-facing interface where customers can submit inquiries, create service tickets, book repairs, track service status, view repair history, and receive notifications.

**Job Order** A formal service record generated from a booking or ticket that contains detailed information about the assigned technician, repair status, service type, and completion timeline.

**Predictive Maintenance** A technique to forecast hardware failures using analytics.

**Repair Tracking** A system function that monitors the progress of repair jobs from initiation to completion, allowing both customers and staff to view real-time status updates.

**Technician Portal** A dedicated interface that allows technicians to view assigned job orders, update repair status, manage availability, and record service notes.

**Ticketing System** A platform to manage and respond to customer service inquiries.

# Chapter II

## Research Design and Methodology

### Design of Software, System, Product or Process

SmartLink will be developed using a three-tier software architecture to enhance the efficiency of the computer services booking system. This architecture consists of three layers: the Presentation Layer, which utilizes HTML, CSS, and JavaScript for the Customer Portal and Staff Dashboard; the Application Layer, responsible for business logic like user authentication, booking validation, scheduling, and technician assignment; and the Data Layer, which employs MySQL for storing and retrieving customer data and booking history. This structure ensures a clear separation of concerns and prevents direct database interactions.

## Software Architecture

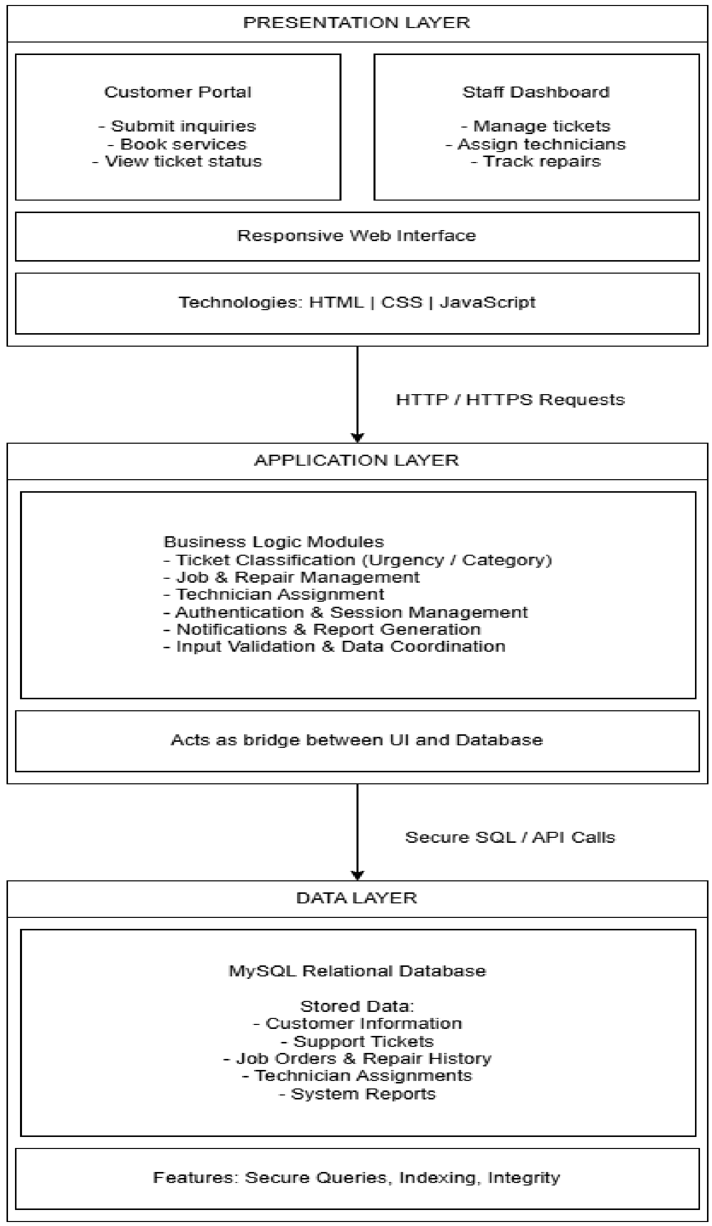



***Figure No. 2:*** *Three Tier Software Architecture*

The SmartLink system is designed using a three-tier software architecture to support an efficient and scalable computer services booking system. This architectural pattern separates the system into three independent but interconnected layers: the Presentation Layer, Application Layer, and Data Layer. The separation of concerns improves maintainability, security, and flexibility while allowing each layer to evolve independently.

### Presentation Layer

The **Presentation Layer** represents the top tier of the system and serves as the primary point of interaction between users and the SmartLink platform. It is accessed through web browsers on both desktop and mobile devices and is developed using standard web technologies such as **HTML, CSS, and JavaScript**.

This layer provides two main interfaces:

- The **Customer Portal**, where customers can browse available computer services, submit service bookings, select preferred schedules, and track the status of their requests.
- The **Staff Dashboard**, which allows staff and administrators to view bookings, manage schedules, assign technicians, and update service progress.

The presentation layer is responsible solely for handling user inputs, displaying system responses, and ensuring a responsive and user-friendly interface. It does not contain business logic; instead, it forwards user requests to the application layer via secure HTTP/HTTPS communication.

### Application Layer

The **Application Layer** forms the core of the SmartLink system and contains all **business logic and processing rules**. It acts as an intermediary between the user interface and the database, ensuring that system operations follow defined rules and workflows.

Key responsibilities of this layer include:

- Processing service booking requests and validating service availability
- Managing schedules and assigning technicians to bookings
- Classifying and updating booking or job statuses
- Handling user authentication and role-based access control
- Managing notifications such as booking confirmations and status updates
- Validating user inputs and coordinating data exchange between system modules

By centralizing business logic in this layer, SmartLink ensures consistency in system behavior while allowing the presentation and data layers to remain loosely coupled.

### Data Layer

The **Data Layer** is the lowest tier of the architecture and is responsible for the **storage, retrieval, and management of system data**. SmartLink utilizes a **MySQL relational database** to maintain structured and secure records.

The data stored in this layer includes:

- Customer and staff information
- Available computer services
- Service bookings and schedules

- Technician assignments
- Service and repair history
- System-generated reports

Data access is performed through secure SQL queries issued by the application layer. Proper database indexing and integrity constraints are implemented to ensure efficient performance, data accuracy, and reliability.

### Overall Architecture Flow

User interactions originate in the **Presentation Layer**, where requests are sent to the **Application Layer** for processing. The application layer applies business rules and communicates with the **Data Layer** to store or retrieve information. Processed results are then returned to the presentation layer for display to users. This structured flow ensures a secure, maintainable, and scalable system.

### Architectural Benefits

The adoption of a three-tier architecture provides several benefits to SmartLink:

- **Separation of concerns**, simplifying development and maintenance
- **Improved security**, as direct access to the database is restricted
- **Scalability**, allowing individual layers to be enhanced independently
- **Flexibility for future enhancements**, such as mobile apps or additional services

Data Flow Diagram Level 0

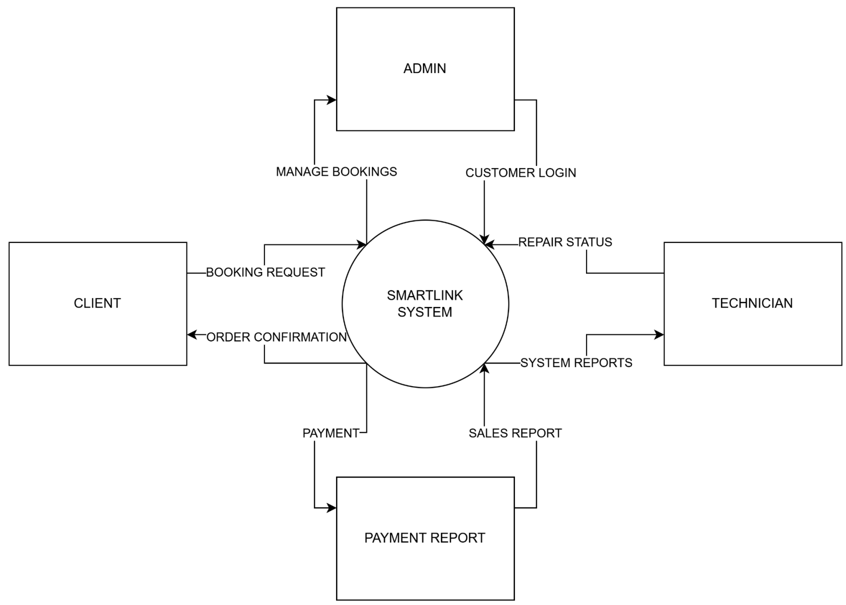



***Figure No. 3:*** *Data Flow Diagram - Level 0 (Gane & Sarson Notation, 1979)*

The Level 0 Context Diagram provides a high-level overview of the SmartLink System, representing the entire platform as a single central process. It maps out the boundary of the system by defining its interactions with the external entities: the Client, Admin, Technician, and the Payment Report entity.

The data flows between the central system and these external entities are organized as follows:

**1. Client (Customer/User)**

**Booking Request - System**: The client initiates a service request by submitting booking details to the system.

**System - Order Confirmation**: The system processes the booking and returns an order confirmation back to the client.

**2. Admin**

**Manage Bookings - System**: The admin interacts with the system to review, update, or assign incoming bookings.

**Admin - Customer Login**: Admin manage and process customer credentials or verification actions for system access.

**3. Technician**

**Repair Status - System**: The technician inputs progress updates, notes, and repair milestones.

**System - System Reports**: The system outputs relevant technical metrics or job details to help the technician monitor their assigned tasks.

**4. Payment Report**

**System - Payment:** Financial data or invoice confirmation is routed from the central system.

**Sales Report - System**: Financial and transaction statistics are fed back into the system to keep business records accurate.

This simplified diagram illustrates how information is routed directly to and from the central database and business modules, keeping the user interface and database secure.

**Data Flow Diagram Level 1**

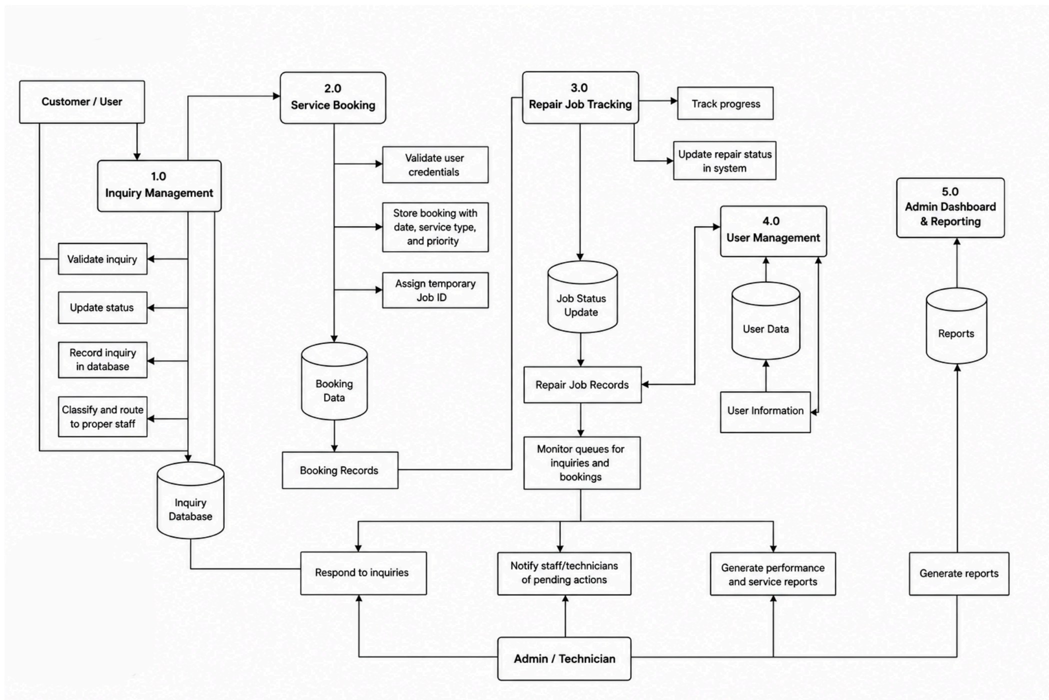



***Figure No. 4:*** *Data Flow Diagram - Level 1 (Yourdon & DeMarco, 1980)*

This diagram illustrates how the system operates through five (5) core processes and its interactions with the external entities, namely the Customer/User and the Administrator/Technician. It provides a detailed view of how data flows within the system and how information is processed, stored, and retrieved.

**External Entities:**

**Customer/User**

- Initiates inquiries, Books services (repairs, IT support), Receives repair status updates

**Administrator/Technician**

- Manages user accounts, responds to inquiries, updates repair job statuses, generates reports

**Primary Processes:**

**Inquiry Management**

- Receives inquiries submitted by customers, Validates and records inquiries in the database, Classifies and routes concerns to the appropriate support staff, Provides inquiry status updates to customers

**Service Booking**

- Handles customer requests for repair or IT services, validates user credentials, stores booking details such as service type, preferred date, and priority, assigns a temporary job ID and confirms booking status

**Repair Job Tracking**

- Tracks the progress of ongoing repair jobs, Updates job status (Ongoing, Completed, Cancelled), Monitors queues for inquiries and bookings, Notifies stakeholders of job status updates.

**User Management**

- Maintains customer and staff profiles, Stores user information and service history, manages authentication and access control, Responds to user-related inquiries

**Admin Dashboard and Reporting**

- Allows administrators to monitor overall system performance, generates service, repair, and performance reports, oversees booking and inquiry queues for efficient operations

**Data Stores:**

- **User Information** – Stores personal details, login credentials, and user profiles, **Booking Records** – Contains service booking information and preferences, **Inquiry Databas**e – Holds submitted inquiries and their current status, **Repair Job Record**s – Logs repair jobs, assignments, and progress updates, **Service History** – Archives completed services and repair records for customers.

  Here at this level more activities of the system appear in detail.

- The first one is Inquiry Management, would receive user inquiries and forward it/generate referrals for the appropriate support personnel.
- The second would be Service Booking – it allows users to book repair or IT services.
- Third is Repair Job Tracking - it manages and keeps updating the ongoing repair service with its status.
- User Management - encompasses keeping records of users along with their respective service history.
- Admin Dashboard - provides Avenue for administrators to generate reports and manage queues.

Such Data Stores as User Information, Job Orders, and Service History form points where information is stored and retrieved by various processes. The arrows in the diagram represent data movements between users, processes, and data stores. Thus, overall, the DFD identifies the primary user interactions, ensuring that data is processed and routed in such a way as to provide efficient and responsive services within SmartLink.

**Data Flow Diagram Level 2**

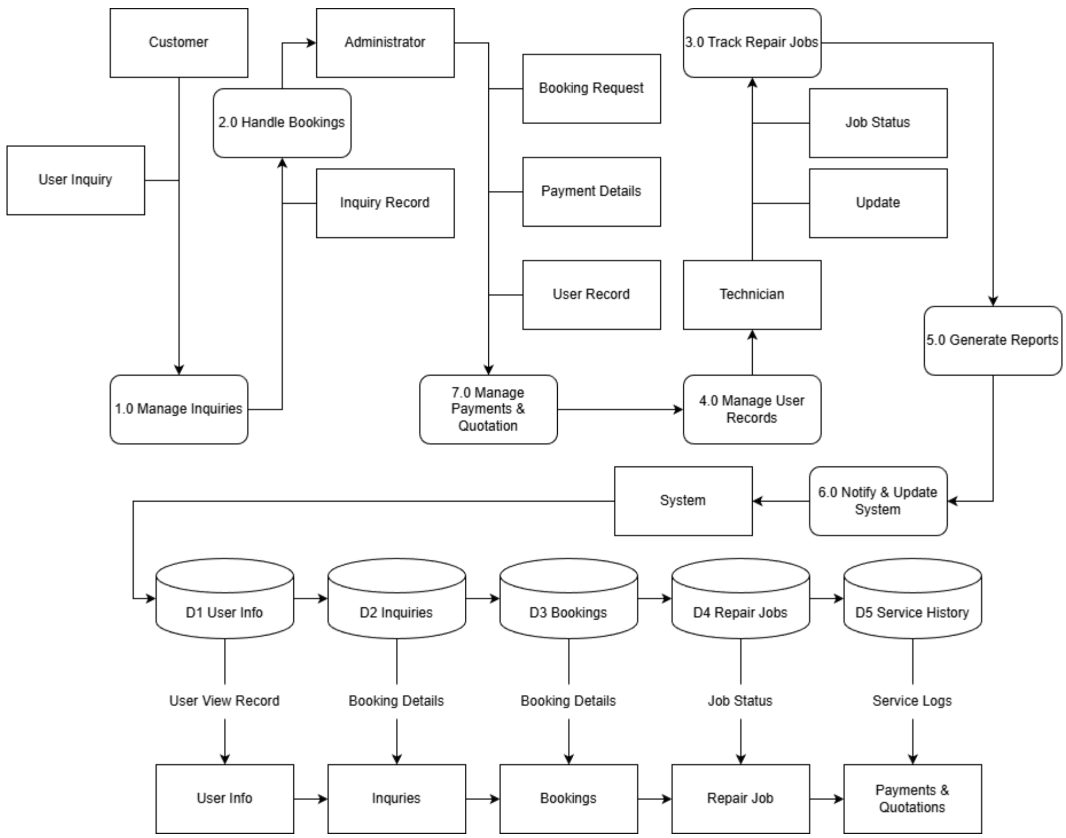



***Figure No. 5*** *Data Flow Diagram - Level 2 (Yourdon, 1988)*

The Level 2 Data Flow Diagram (DFD) for the SmartLink system provides a more detailed breakdown of core processes in the platform:

- **Inquiry Management:** Receives customer inquiries and refers them to appropriate support personnel. It updates the inquiry status based on action taken.

- **Service Booking:** Allows customers to book services and stores booking preferences and schedules.
- **Repair Job Tracking:** Handles assignment of jobs to technicians, tracks job progress and updates status.
- **User Management:** Maintains user records and their service histories, login credentials, and roles.
- **Admin Dashboard:** Provides access for administrators to generate reports and monitor the platform.

**Data Stores include:** User Info, Inquiries, Bookings, Repair Jobs, and Service History.

**External Entities**

| **External Entity** | **Description** |
|---|---|
| Customer | Initiates service inquiries, bookings, and checks job statuses. Also provides feedback and receives updates. |
| Technician | Handles assigned repair jobs, updates job status, and provides repair notes. |
| Administrator | Manages user records, assigns technicians, oversees the dashboard, and generates reports. |
| System | Sends notifications, stores data, and logs activities (acts as a system |

| | |
|---|---|
| | component providing internal automation). |

**Processes**

| Process No. | Process Name | Description |
|---|---|---|
| 1.0 | Manage Inquiries | Receives customer inquiries, logs them, and forwards them to the admin or technician. |
| 2.0 | Handle Bookings | Accepts and schedules service bookings. |
| 3.0 | Track Repair Jobs | Assigns technicians, updates job progress, and logs status changes. |
| 4.0 | Manage User Records | Adds and updates customer/technician/admin records. |
| 5.0 | Generate Reports | Prepares automated reports based on system logs and service records. |
| 6.0 | Notify and Update System | Sends email notifications and updates customers/admins. |
| 7.0 | Manage Payments and Quotation | Handles service quotations and payment confirmation details. |

**Software Development Cycle**

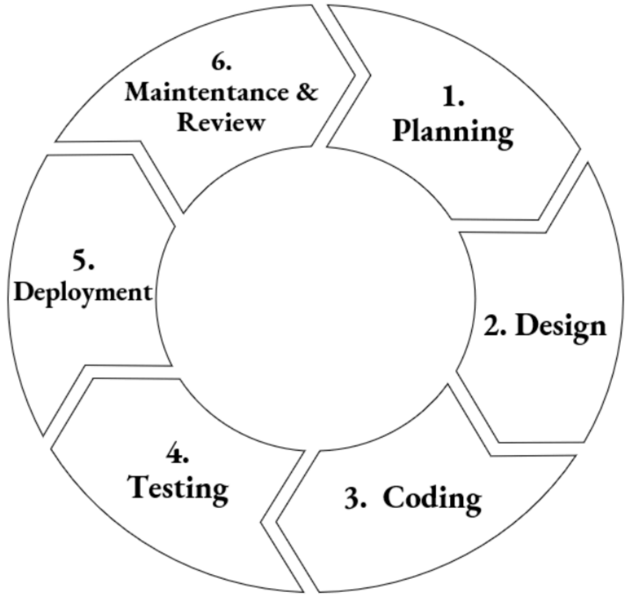



***Figure No. 6***. *Software Development Cycle (Royce, W. W. 1970)*

**The Software Development Life Cycle (SDLC)** is a structured process that guides the development of high-quality software in a systematic and cost-effective way. The image above illustrates the SDLC as a continuous, circular process consisting of six key phases, ensuring that software is developed efficiently, meets user requirements, and remains maintainable over time.

**Planning -** This initial phase involves gathering requirements, defining the project scope, setting objectives, and identifying resources. Effective planning ensures that the development process aligns with business goals and minimizes risks.

**Design -** In this stage, the software's architecture and technical design are developed based on the gathered requirements. This includes defining system components, interfaces, data models, and security features to create a blueprint for development.

**Coding (Development) -** During the coding phase, developers translate the design into actual software code using the chosen programming languages and tools. It is essential that code follows best practices, standards, and is well-documented to ensure future maintainability.

**Testing -** Once coding is completed, the software undergoes rigorous testing to identify defects, security vulnerabilities, and performance issues. Testing ensures the system functions as intended and meets the required quality standards.

**Deployment -** After successful testing, the software is deployed to a live environment where end-users can access and use it. This may involve installation, configuration, and providing user training if necessary.

**Maintenance & Review -** Software maintenance involves fixing bugs, updating features, and ensuring the system remains functional and secure over time. Regular reviews help identify opportunities for improvement and ensure the software continues to meet evolving user needs. The SDLC provides a clear, repeatable process that enhances software quality, improves project management, and reduces development risks. By following these stages, organizations can deliver reliable, efficient, and user-friendly software solutions.

**System Flow Chart**

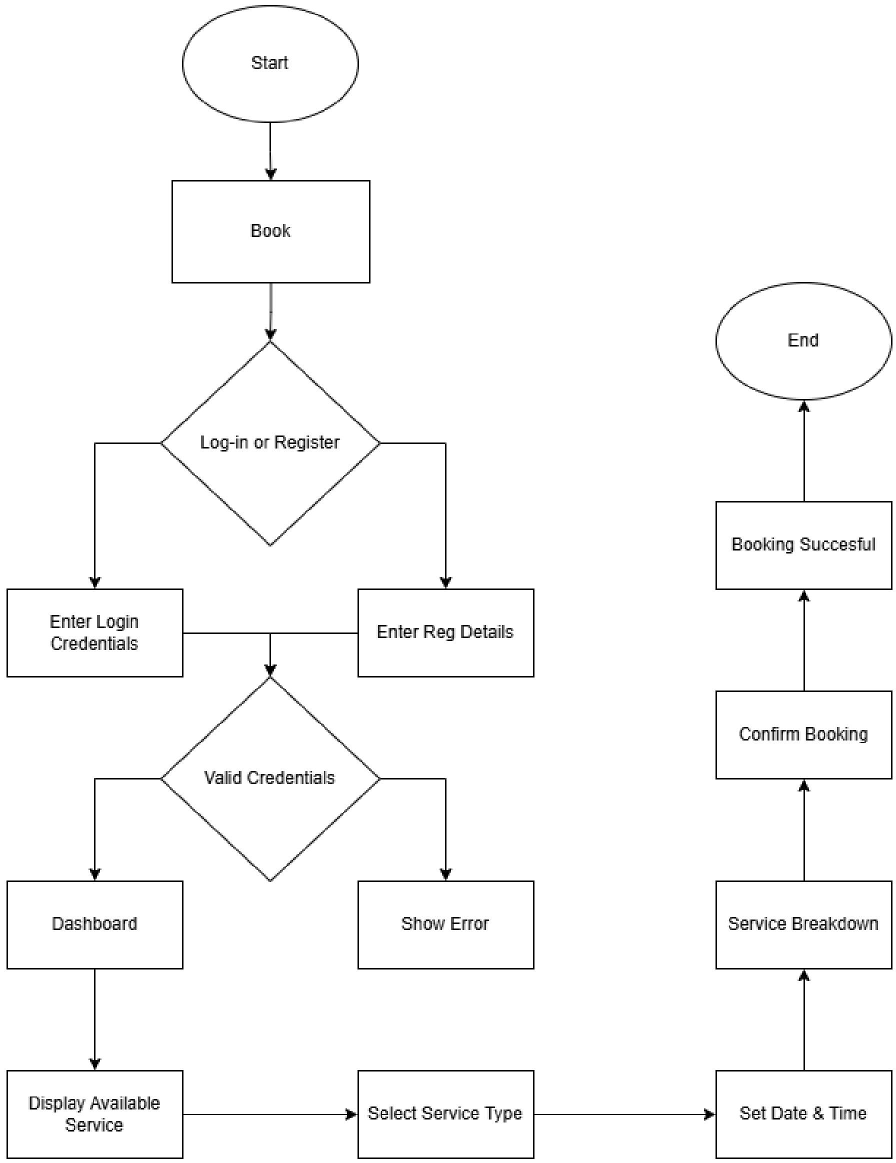



***Figure No. 7:*** *Customer System Flowchart*

**Start**

The process begins when a customer accesses the SMFP website.

**Business Website**

The customer visits the company's website to explore available services.

**Customer Inquiry (Left Path)**

**Submit Inquiry**

The customer submits a question or concern, which may not be a formal service request.

**Create Ticket**

A new service ticket is automatically created to document the inquiry.

**Update Ticket Status**

The system or staff updates the status of the inquiry (e.g., “In Review”, “Resolved”).

**End Inquiry Flow**

The process ends once the inquiry has been addressed.

**Booking and Service Request Flow (Middle and Right Paths)**

**Book Repair Service**

The customer chooses to book a repair or service request.

**Decision: Log In or Register?**

**Log In:** The customer enters their login credentials.

**Decision: Are credentials valid?**

- **Yes → Dashboard:** The system shows available services.
- **No → Retry Login:** Customer is prompted to re-enter credentials.

**Register:** The system collects required information to create a new account.

After successful registration, the customer is redirected to the dashboard.

**Select Service**

The user selects a service type, preferred date, and time.

**Service Fee & Details**

The system shows a breakdown of fees, address, and contact information. A disclosure notes that the total bill may change if additional services are needed.

**Decision: Submit Ticket?**

> **Yes → Create Ticket:** The booking request is recorded as a formal ticket.
>
> **No → Reschedule / Drop Service:** The customer can reschedule or cancel the request.

**Assign Technician**

A technician is automatically assigned to the service order.

**Decision: Set Appointment Date?**

> **Yes → Proceed with Service**
>
> **No → Halt / Return to Reschedule**

**Status & Feedback**

**Request Status Update:** Customers can check updates on their booking.

**Update Repair Status:** The system or technician updates the repair status after final service actions.

**Booking Completed**

The system confirms the successful completion of the booking and updates all relevant records.

**End Booking Flow**

The process concludes with the service request fully processed, whether completed, rescheduled, or canceled.

**Payment System Flowchart**

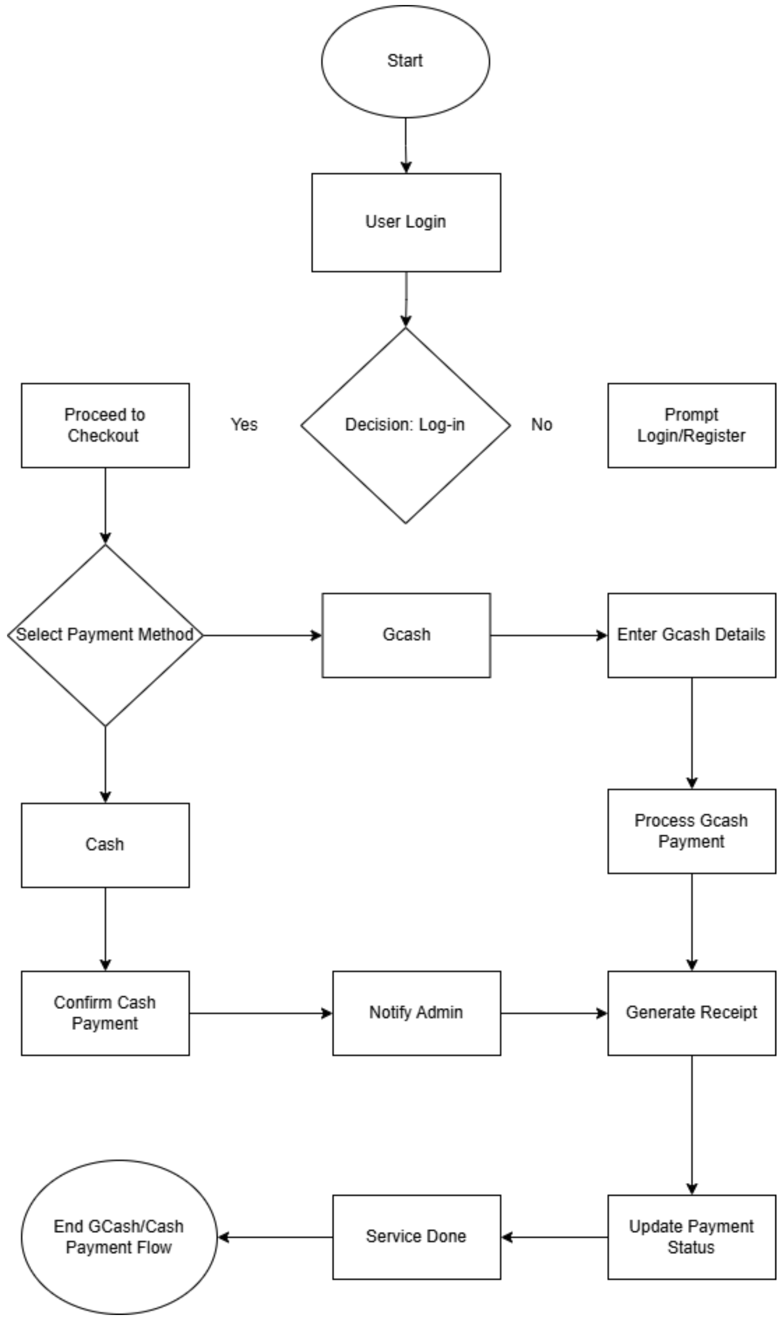



***Figure No. 8:*** *Payment System Flowchart*

**Start**

The process begins when the user proceeds to checkout after booking a service.

**User Login / Access**

The system checks if the user is logged in.

**Decision:** Logged in?

**Yes → Proceed to Checkout**

**No → Prompt Login or Register**

**Select Payment Method**

The user chooses the preferred payment method: **Cash** or **GCash**.

### Cash Payment Path

**Confirm Cash Payment**

The user confirms that they will pay in cash upon service completion.

**Generate Order / Receipt**

The system generates an order summary and receipt.

**Notify Admin**

The system sends a notification to the admin about the pending cash payment.

**Service Completion**

Cash is collected when the service is delivered.

**Update Payment Status**

The admin or system updates the order status to "Paid."

**End Cash Payment Flow**

The payment process concludes successfully.

### GCash Payment Path

**Scan GCash QR**

The user scans the QR and inputs required information (GCash number, amount, etc.).

**Process GCash Payment**

The technician validates the details and confirms the payment.

**Decision: Payment Successful?**

**Yes → Generate Receipt**

**No → Retry Payment**

**Notify Admin**

The system automatically notifies the admin that payment was received.

**Update Payment Status**

The system updates the order status to "Paid."

**End GCash Payment Flow**

The payment process concludes successfully.

Admin System Flowchart

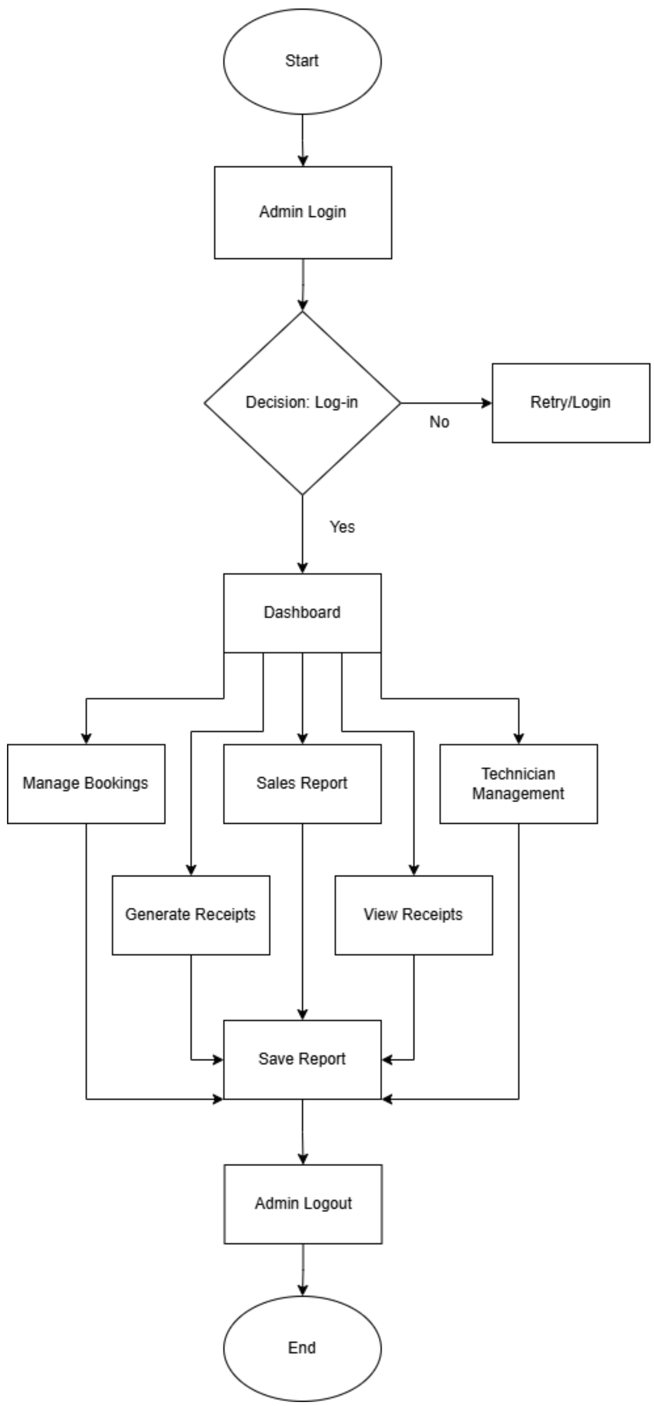


***Figure No. 9:*** *Admin System Flowchart*

**Start**

The process begins when the admin accesses the SMFP system.

**Admin Login**

The admin logs in through the Admin Login screen.

**Decision: Valid Credentials?**

**Yes → Dashboard:** The admin is redirected to the dashboard.

**No → Retry Login:** An error message appears, prompting the admin to re-enter credentials.

**Dashboard Modules**

**Manage Booking**

Enables the admin to manage customer orders.

Process transactions, generate receipts, and issue order numbers.

**Technician Management**

Allows the admin to view existing technicians.

Provides functionality to add new technicians.

**Analytics**

Provides insights into shop performance.

Allows viewing sales forecasts, analyzing customer ordering patterns, and identifying peak booking times.

Supports data-driven decisions to enhance operations and optimize sales strategies.

**Sales Report**

Allows the admin to generate and review sales summaries.

Ensures accurate tracking of transactions and overall business performance.

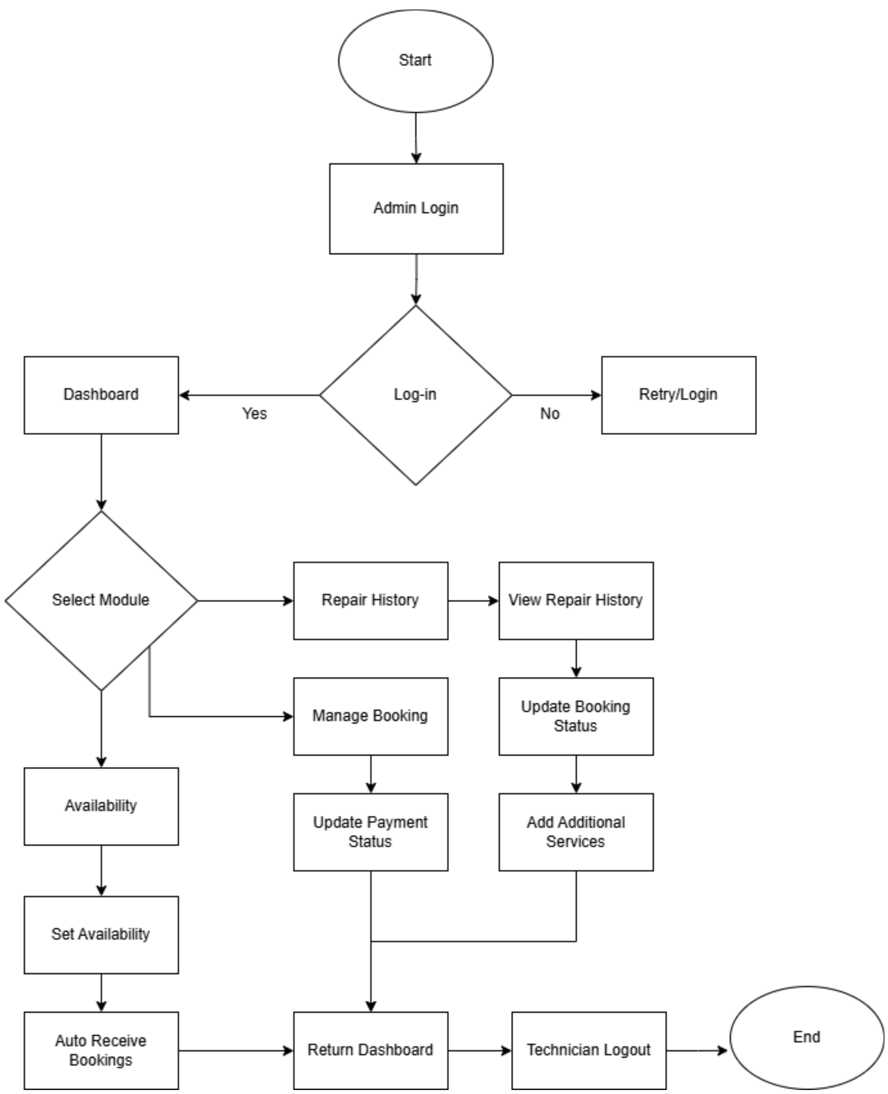
Provides real-time information on sales and the number of customer bookings.

**Admin Logout**

The admin clicks the logout button to securely exit the system.

**End**

The admin session concludes, ensuring all operations are complete and the system is securely closed.




Technician System Flowchart

***Figure No. 10:*** *Technician System Flowchart*

**Start**

The process begins when the technician accesses the SMFP system.

**Technician Login**

The technician logs in through the **Technician Login** screen by entering valid credentials.

**Decision: Valid Credentials?**

**Yes → Dashboard**

The system redirects the technician to the dashboard.

**No → Retry Login**

An error message is displayed, prompting the technician to re-enter login credentials.

**Dashboard Modules**

**Availability**

The technician manually sets their availability schedule.

The system uses the availability information to automatically assign service bookings based on the technician's specialization.

**Manage Booking**

The technician views assigned service bookings.

The technician updates the booking status.

The technician updates the payment status.

The technician adds additional services when applicable.

**Repair History**

The system displays the technician's complete repair history.

The technician reviews past and completed service records.

**Technician Logout**

The technician clicks the **Logout** button to securely exit the system.

**End**

The technician session concludes, ensuring all operations are completed and the system is securely closed.

**Entity Relationship Diagram**

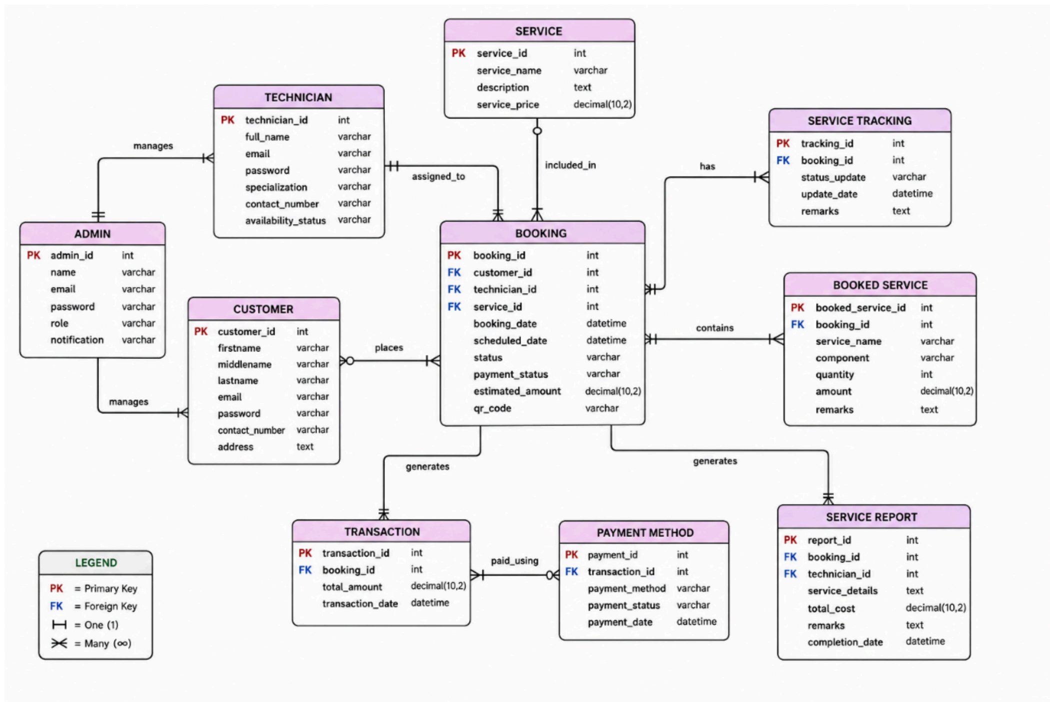



***Figure No. 11:*** *SmartLink's Entity Relationship Diagram*

The Entity Relationship Diagram (ERD) for the SmartLink platform outlines the structure of the database and how entities interact with one another in the system.

**1. Users** This entity consists of the people who access the platform. User data incorporates name, e-mail, contact number, and role (admin, customer, or technician). A user can generate multiple tickets and job orders.

**2. Tickets** This entity represents issues reported by customers. Each ticket has a unique ID, issue description, urgency level, status, and is linked with some user. Tickets are a means of tracking customer inquiries and problems to be resolved with support.

**3. Job Orders** This entity keeps detailed repair or service requests. A job order is connected to a user and may be derived from a ticket. It contains information such as technician assignment, job description, status, and timestamps for tracking.

**4. Services** This entity holds a list of IT services that the company has on offer (for example, hardware repair or software installation). Job orders are attached to services to help classify each job in terms of its type.

**5. Technicians** This entity has technician particulars including their specialty and contact information. Technicians who are listed in the system can be assigned to the Job Order based on their particular skillset.

**6. Repair History** It includes the log of the job orders on which jobs have been completed and very important service notes as well as customer feedback and timestamps. This keeps a source of historical information for reference and for service planning in the future.

**8. Inventor**y This optional entity shows components/parts used during job orders. Each record refers to the names of items, quantity and detailed information about using it, thus ensuring transparency and monitoring for supply.

A user can have many tickets and job orders.  Technicians can assign one to multiple job orders. A job order can be the association of a service. The tickets and the job orders are both linked to the users; the job orders can originate from tickets. Notifications are associated with the users and the events, the ticket/job order updates.

Thus, this ERD lays down a strong logical framework for a structured and scalable repair and support management system.

The following tables represent the database structure for SmartLink: A Unified Platform for IT Services and Customer Support.

## Data Dictionary

### Customer Table

| Field Name | Data Type | Description |
|---|---|---|
| user_id | INT (PK) | Unique identifier for each user |
| name | VARCHAR | Full name of the user |
| email | VARCHAR | User's email (for login/communication) |
| password | VARCHAR | Hashed password |
| user_type | ENUM | 'Customer', 'Technician', 'Admin' |
| date_registered | DATE | Date the user account was created |

### Inquiry Table

| Field Name | Data Type | Description |
|---|---|---|
| inquiry_id | INT (PK) | Unique ID for each customer inquiry |
| user_id | INT (FK) | Refers to the user who submitted inquiry |
| message | TEXT | Customer's inquiry or concern |
| date_submitted | DATETIME | Timestamp of when the inquiry was made |
| status | ENUM | 'Open', 'Referred', 'Resolved' |

### Booking Table

| Field Name | Data Type | Description |
|---|---|---|
| booking_id | INT (PK) | Unique ID for each booking |
| user_id | INT (FK) | Customer who booked the service |
| service_type | VARCHAR | Type of service booked (e.g., Repair, IT) |
| preferred_date | DATE | Customer's preferred service date |

| | | |
|---|---|---|
| status | ENUM | 'Pending', 'Confirmed', 'Completed' |

**Repair Job Table**

| **Field Name** | **Data Type** | **Description** |
|---|---|---|
| job_id | INT (PK) | Unique ID for the repair job |
| booking_id | INT (FK) | Booking related to the job |
| technician_id | INT (FK) | Assigned technician |
| job_status | ENUM | 'Ongoing', 'Completed', 'Cancelled' |
| diagnosis | TEXT | Issue found |
| repair_notes | TEXT | Notes from technician |
| completion_date | DATE | Date job was completed |

**Requirements Analysis**

Requirements analysis for SmartLink: A Unified Platform for IT Services and Customer Support aims at understanding the specific needs for the client company SMFP and end-users (customers or technicians). The analysis makes sure that the system that will be developed touches on the identified problem appropriately, boosts efficiency, and improves customer service delivery.

The analysis was based on interviews with key personnel from SMFP, namely the IT technician, admin staff, and frequent customers. Besides this, some observations were conducted regarding current job order tracking practices and existing inefficiencies and user frustrations with the available communication modes.

**Functional Requirements:**

The system is to allow customers to create, track, and update service tickets online. Admin must assign tickets to technicians and monitor their progress. The customers will have notifications in real-time regarding the status of their requests. Also, there should be a booking system for scheduling on-site or in-store services. There must be a searchable repair history for every client in the database. They should also set up a QR code system for easy tracking of job order statuses.

**Not-Functional Requirements Specified by the User:**

The system should be web as well as mobile friendly. It should have its data privacy and must be in accordance with the basic data protection policy. The system should be scalable in terms of more user additions and service branches in the future.

**User Requirements:**

The user would like for its online experience to be more professional compared to Facebook chat. It should also have much better transparency in repairs updates with clearer pricing. The technician will manage multiple ongoing repair jobs at a time with the help of the centralized dashboard. The admin will thus need to have automated report generation and predictive insights for better operation.

These particular requirements analysis forms the basis for the next stage, which is designing and developing the required system so that each user role and model functional objective is properly considered in the final implementation.

**Requirements Documentation**

The Requirements Documentation is the formalization of the user needs that were gathered in the analysis phase. It is considered a reference for developers, stakeholders, and testers throughout the system development lifecycle of SmartLink. The documentation consists of four types of requirements: functional requirements, non-functional requirements, user requirements, and system requirements.

**Functional Requirements** - These describe what the system is required to do to fulfill its users' needs. **Ticketing System** that customers may create and submit service tickets online.

Tickets are categorized based on urgency, type of service (hardware, software, consultation), and location. The admin and technician roles are responsible for updating the status of a ticket (Received, In Progress, Completed). Updates regarding every change in status are sent to customers via email, notification.

**Repair Management Module** that admins assign technicians to repair jobs. The technician logs progress by updating status per job.

**Inventory & Records** that tracks parts and inventory used for each repair. History of all service requests by customers is maintained and searchable.

Customer Portal that shows real-time status of ongoing requests. Users are allowed to book appointments and view previous accounts of repair history. Downloading invoices or receipts is available. Advice for simple troubleshooting. Walk the user through the ticket creation process.

**Admin Dashboard** that provides reports on repair timelines, technician performance, and customer satisfaction.

**Non-Functional Requirements** These are the criteria for judging the performance and acceptability of the system.

**Security** that uses role-based access control (Admin, Technician, Customer). All user and ticket data must be encrypted at rest and during transmission.

**Performance** for web pages must load in 2-3 seconds. Supposedly, the system should sustain at least 200 simultaneous users.

**Reliability** that the system must guarantee 99.9% uptime. Backup and restore features must be implemented.

Usability that the interface should be intuitive and requires minimal training for end users. The design should be responsive in layout for mobile.

**Maintainability** that the codebase should be modular and sufficiently commented, allowing for easy updates.

**User Requirements** are the expectations and needs from different user groups:

**Customers** will have easy navigation and clear steps to book repairs. Transparent status updates.

**Technicians** that task dashboard viewing assigned jobs and deadlines. Ability to log progress and attach notes or photos.

**Administrators** that can overview system usage and analysis. Manage users of the system, content (FAQs, blog), and reports.

**System Requirements refer** to the technical requirements for smooth operation of the system. Hardware Requirements, Web Server: Intel i7 Processor, 16GB RAM, 1TB SSD, Client Devices: Mobile phones, tablets, or PCs with browsing facilities

**Software Requirements** Frontend: HTML5, CSS3, Javascript (React.js), Backend: Node.js or PHP (Laravel), Database: MySQL or PostgreSQL, Hosting: Cloud-based (e.g., AWS, Firebase, or Hostinger)

This document aims to ensure that the development team has at least some baseline common working knowledge of what the system should accomplish, how it will perform, and how it will be evaluated. It reduces the risks of scope creep and mismatched expectations during the whole project life cycle.

**Development and Pre-Testing Procedures**

This phase includes development and pre-testing activities where the system is built as per the given approved design. It is also ensured that the above-defined modules are entirely functioning prior to undergoing the full deployment phase. The process is methodical and systematic following a clearly defined Software Development Life Cycle (SDLC) for ensuring that the system is accurate, reliable, and usable.

**Development Procedures**

1. Development Environment Setup

- Configured tools such as Visual Studio Code, MySQL Workbench, and Postman. Initialized the repository using Git and GitHub as a version controller for the system.

2. Database Design and Implementation

- Live relational database creation was achieved through MySQL from the ERD (Entity Relationship Diagram). Some data entries were inserted for query tests and report generation.

3. Frontend Development

- Responsive UI using HTML, CSS, and JavaScript. Customer and admin views such as dashboards, ticket forms, trackers for repairs, and inquiry modules.

4. Backend Development

- Used Node.js (or PHP-Laravel) in developing server-side logic. API endpoints were created that would manage requests for ticketing, login/authentication, repair management

**Pre-Testing Procedures**

The internal developers performed some pre-test procedures to validate core functionalities before releasing the system for proper user testing.

1. Unit Testing

- Testing was performed for each module (Ticket Creation, Repair Update, Admin Dashboard) separately to find bugs and logical programming issues.

2. Integration Testing

- It tested the interaction amongst the modules so that data could flow well (e.g. notifying module response when a ticket is updated).

3. UI/UX Testing

- UI/UX tests were being done by testers on various devices (mobile, tablet, desktop) to measure interface responsiveness and ease of use.

4. Security and Access Testing

- The access was verified to authentication checks. Login checks were performed along with password encryption and session management.

5. Backup and Recovery Simulation

- Mock failure scenarios were created for testing database backup and restoration.

This development and pre-testing process ensured that the SmartLink application could be implemented in a variety of settings, but with the belief that it was very usable, secure, and reliable in the ability of its pilot deployment.

**Implementation Plan (Infrastructure/ Deployment)**

The platform will be deployed on a cloud server (e.g., AWS or Hostinger), accessible via web browser. It will use HTTPS, daily backups, and role-based access.

**A. Development Environment**

| **Component** | **Tools / Technology Used** |
|---|---|
| Frontend | HTML, CSS, JavaScript, Bootstrap |
| Backend | PHP (Laravel or Core PHP), or Python (Flask/Django) |
| Database | MySQL or PostgreSQL |
| Version Control | Git, GitHub |
| Text Editor/IDE | Visual Studio Code |
| Design Tools | Figma / Canva (for UI mockups) |
| Local Server Testing | XAMPP / WAMP / Localhost setup |

**B. System Module to be Implemented**

Customer Portal, Inquiry Form, Booking Form, Repair Status Tracker (with Ticket ID), Admin Dashboard, Job Order Management, Customer Records Management, Service Request Logs, Status Update Panel, Repair Technician Portal, Assigned Jobs Overview, Status Update Submission, Notes Upload for Completed Jobs, Notifications System, Email (Optional if feasible) for ticket updates, Web notifications on dashboard

**C. Deployment Infrastructure**

| **Stage** | **Details** |
|---|---|

| | |
|---|---|
| Web Hosting | Free: InfinityFree, 000WebHost, or Paid: Hostinger |
| Domain Name | Free: Freenom (.tk, .ml) or Paid: GoDaddy, Namecheap |
| Database Hosting | PhpMyAdmin (via hosting CPanel) |
| Deployment Tool | Manual file upload via FTP or CPanel |

**D. Milestone Timeline**

| Phase | Description | Target Duration |
|---|---|---|
| Planning | Finalize features, diagrams, and scope | Week 1 |
| System Design | UI wireframes and database schema | Week 2 |
| Frontend Dev | Build main interface, navigation, forms | Week 3 – 4 |
| Backend Dev | Connect to DB, handle CRUD operations | Week 4 – 6 |
| Testing Phase | Functional, usability, and integration testing | Week 6 – 7 |
| Deployment | Host website online, test with dummy data | Week 8 |
| User Training | Provide demo and manual to client | Week 9 (if required) |
| Final Documentation | Finalize paper with screenshots and results | Week 10 |

**E. Maintenance Plan**

**Bug Fixing** that is ongoing after deployment based on user feedback.

**Feature Expansion** that prepares for future improvements like adding search filters, mobile responsiveness, etc.

**Backup** that is manual database backup weekly.

**Security** that uses form validation, sanitization to prevent SQL injection.

**Statistical Treatment of Data**

**Data Gathering**

To evaluate the overall performance and usability of the developed Ticketing and Repair Management System with an Integrated Business Website for SMFP. The survey will be administered to customers, technicians, and the business owner after they have used the system. It will contain quantitative questions, such as Likert-scale items to measure usability, functionality, efficiency, satisfaction, and system performance. In addition, system-generated data, such as ticket processing time and repair completion time, will be collected to support the survey results. This mixed-method approach will allow the researchers to assess the system's effectiveness in improving customer interaction, service tracking, and overall business operations.

**Settings of the Study**

The study will be conducted at SMFP, a repair and IT services company, where the Ticketing and Repair Management System with an Integrated Business Website was developed and implemented. The setting involved the actual business environment of SMFP, including its service operations, customer interactions, and repair management processes.

## Subject of the Study

The study will focus on customers, technicians, and the business owner of SMFP who used the Ticketing and Repair Management System with an Integrated Business Website. Customers provided feedback on usability and service experience, technicians evaluated the system's support for repair and ticket management, and the business owner assessed its use in monitoring operations. These participants were chosen because they directly used the system.

## Sampling Technique

The study used purposive sampling, selecting participants based on their direct use of the Ticketing and Repair Management System with an Integrated Business Website. The participants included customers who submitted service tickets, technicians who managed repairs and updated ticket statuses, and the business owner who monitored operations and records. This method ensured that only those with experience using the system were included, providing relevant feedback for its evaluation.

## Statistical Treatment of Data

The data collected from the surveys and system metrics will be analyzed using quantitative methods. Quantitative data, such as satisfaction ratings and system performance metrics, will be summarized using frequency counts, percentages, and mean scores. This approach will allow the researchers to evaluate the system's usability, efficiency, and effectiveness based on the feedback of customers, technicians, and the business owner.

**Simple Percentage**

One of the most commonly used methods for presenting statistics is percentage analysis. In research tabulation, the percentages reflect the respondents' answers and show the proportion of results within each category.

The formula applied was:

$$P = \left(\frac{f}{n}\right) x\ 100$$

Where:

P = percentage of respondents

f = frequency for each rating

n = total number of respondents

**Weighted Mean**

This will be used to calculate an average score for each characteristic across all responses, providing a summary of the overall participants' ratings.

The Formula Applied was:

$$\underline{X} = \frac{\Sigma f x}{\Sigma_f}$$

Where:

$\underline{X}$ – is the weighted mean

x – is the rate value (1-5)

f – is the number of the respondents

The weighted average of the values recorded for each evaluation question will be analyzed based on the scale presented in the table below.

| Range – Value | Description |
| --- | --- |
| 4.50 – 5.00 | Strongly Agree/ Highly Acceptable |
| 3.50 – 4.49 | Agree / Very Acceptable |
| 2.50 – 3.49 | Neutral / Moderately Acceptable |
| 1.50 – 2.49 | Disasgree / Fairly Acceptable |
| 1.00 – 1.49 | Strongly Disagree / Not Acceptable |

**Standard Deviation**

According to *Statista's Statistics Glossary*, standard deviation is a statistical measure that shows how much the values in a dataset vary from the mean, and it is calculated by taking the square root of the variance, which itself is the average of the squared differences from the mean.

The Formula Applied was:

$$S = \sqrt{\frac{\sum f(x - \underline{X})^2}{\sum f}}$$

Where:

f – frequency of each rating

x – rating value

$\underline{X}$ – weighted mean

$\sum f$ – total number of respondents

**Table: Likert Scale**

| Numerical Rating | Description |
|---|---|
| 5 | Strongly Agree |
| 4 | Agree |
| 3 | Neutral |
| 2 | Disagree |
| 1 | Strongly Disagree |

This table provides a way for the researchers to convert numeric results into a quantitative format, making the findings easier to understand.

# CHAPTER III

## RESULTS AND DISCUSSION

This chapter presents the results of the system evaluation and discusses the findings based on the responses gathered from end-users. The assessment aimed to determine the effectiveness, reliability, usability, security, and overall satisfaction of the developed repair and service management system for SMFP Computer. Data were collected using a structured questionnaire and analyzed through descriptive statistical methods.

A total of thirty-seven respondents participated in the evaluation. The questionnaire utilized a five-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5). The results are organized into seven major evaluation criteria: Functional Suitability, Performance and Responsiveness, Usability and User Experience, Reliability, Security and Data Protection, Accessibility and Compatibility, and Overall Satisfaction.

### System Test Plan

The system test plan was designed to assess the system's functionality, stability, and user acceptance under real operating conditions. Users were allowed to interact with the system by submitting repair requests, monitoring service status, and accessing system features relevant to SMFP Computer's daily operations.

Testing focused on evaluating how effectively the system supports service request submission, repair job management, notification delivery, and user interaction. Feedback from actual users served as the basis for determining the system’s readiness for deployment and operational use.

The procedure outlined below was followed to operate the SMFP Booking system on the user side:

1. The customer initiates the booking process by tapping "Book" on the homepage.
2. The system asks the customer to either log in or register.
3. If Log In is selected, the system requests the customer’s login credentials.
4. After valid credentials are entered, the system proceeds to the dashboard.
5. If “Register” is selected, the system collects the required information to create a new account.
6. After successful registration, the customer is redirected to the dashboard.
7. The system shows the services available.
8. The user selects a service type, and details (Preferred date, and time)
9. The user is redirected to a breakdown of the service fees, along with the address and contact information. A disclosure is also shown indicating the total bill, which is subject to change in case additional services are required.
10. The user booked successfully.

The procedure outlined below was followed to operate the SMFP Booking system on the admin side:

1. The admin accesses the system through the Admin Login screen.
2. After a successful login, the system redirects the admin to the dashboard.

3. From the Dashboard, the admin can select from the following modules:
4. Manage booking that enables the admin to manage customer orders, process transactions, generate receipts, and issue an order number.
5. Technician that enables admin to see the technicians and can also add new technicians.
6. Analytics that the analytics module offers the admin insights into the shop's performance. It enables viewing of sales forecasts, analysis of customer ordering patterns, and identification of peak booking times. This module supports data-driven decision-making to enhance operations and optimize sales strategies.
7. Sales Report allows the admin to generate and review sales summaries. It ensures accurate tracking of transactions and overall business performance, providing real-time information on sales and the number of customer bookings.
8. The admin clicks Admin Logout to securely exit the system.

The procedure outlined below was followed to operate the SMFP Booking system on the technician side:

1. The technician accesses the system through the Technician Login screen.
2. After a successful login, the system redirects the technician to the dashboard.
3. From the Dashboard, the technician can select from the following modules:
4. Availability that the technician can manually set their availability for them to automatically receive booking of their specialization.

5. Manage Booking that the technician can update the booking status and payment status, as well as add any additional services.
6. Repair History that the technician can view their complete repair history.
7. The technician clicks Technician Logout to securely exit the system.

**Table No. 1:** Functional Effectiveness testing procedures

| Test Case | Steps to be taken | Expected Output | Results |
|---|---|---|---|
| **User Registration and Login** | Select "Register," enter valid details (name, email, password), and submit. Select "Login," enter valid credentials, and click submit. | An account is created, and the user is redirected to the dashboard. The user is authenticated and redirected to the dashboard | **Passed** |
| **Service Display and Selection** | Choose available services on the dashboard. Choose the service the user wants to book. | The system displays all available service types. The selected service proceeds to also select their preferred date and time. | **Passed** |
| **Service Breakdown Information** | View the total service fees, along with the input field for address and contact information. | The system sees the total service fees clearly displayed, enter their address in a designated input field, and | **Passed** |

| | | | |
|---|---|---|---|
| | | enter their contact information in a corresponding input field. | |
| **Booking History Dashboard and Transactions** | Select “Booking History” to see all the pending, in progress, completed, and cancelled booking.<br>Select the “Transactions”on the dashboard to view the completed bookings along with their receipt. | The system can clearly display their booking history.<br>The system can clearly display the completed transactions with a button to see the receipt. | **Passed** |
| **Admin Dashboard and Management** | Login as Admin (Manage Booking, Manage Transaction, Sales Report) | Admin can view , generate receipts, accept additional services from technicians, view transactions, and sales reports. | **Passed** |

**Table No. 2:** Usability Testing Procedures

| Test Case | Steps to be taken | Expected Output | Results |
|---|---|---|---|
| **Completeness of Booking Workflow** | Performs a complete booking by starting the system, browsing the services provided, selecting a service, and proceeding to input the full address and contact information of the user for them to submit their booking. | The entire process is seamless and complete with no steps missing. | **Passed** |
| **Automation Features** | Automatically selects a technician of their respective specialization. | The technician is automatically assigned based on their specialization, ensures that the assigned technician | **Passed** |

| | | | |
|---|---|---|---|
| | | matches the required service type, and the user sees in the booking history indicating which technician has been assigned to their booking. | |
| **Completeness of Services Provided** | The users must book a service | The user will see all the provided services (Hardware Upgrade, Software Installation) | **Passed** |
| **Admin Dashboard Usefulness** | The admin logs in and is directed to the dashboard | The buttons and modules offer all the essential management functions. | **Passed** |
| **Completeness of Reports** | View the sales report | The report includes sales totals, number of bookings, and key needed information | **Passed** |

## Statistical Analysis

### 1. Data Collection & Evaluation Scope

The evaluation of the SMFP System followed a structured system test plan focusing on performance, usability, and overall quality. A survey was administered via Google Forms to collect quantitative data from 37 total respondents, which included 32 Stakeholders and 5 IT Experts. The survey specifically targeted the evaluation of the Ticketing and Repair Management System with an Integrated Business Website.

### 2. Statistical Tools Used

To provide a clear, organized summary of the evaluation results, the collected data was processed using three primary statistical methods:

Frequency Counts: Tracking the exact number of times a specific response or rating was chosen.

Simple Percentages: Calculating the proportion of specific responses to determine the frequency distribution of the ratings.

Mean Scores: Computing the weighted mathematical average for each survey item to identify the general consensus of the respondents.

### 3. Purpose & Analytical Outcome

These statistical methods allowed the researchers to objectively measure the system's usability, efficiency, and effectiveness from the distinct perspectives of both standard users and IT experts. By synthesizing the frequency, percentages, and mean scores, the final data presents a comprehensive, easily understandable view of how well the system fulfills its operational goals.

## Implementation and Statistical Results Interpretation

### Implementation

The SMFP Ticketing and Repair Management System was created to enhance customer interaction, service tracking, and business visibility for SMFP, a repair and IT services company. Built with PHP, JavaScript, AJAX, MySQL, and XAMPP, and developed using VS Code, the system includes three main components: a customer portal for submitting and tracking service tickets, a technician portal for updating ticket and payment statuses, and an admin dashboard for managing and overseeing the entire process. Additionally, the system integrates a business website that provides details about the company's services.

The development process focused on creating a user-friendly interface while ensuring the system's core functionalities were effective. AJAX was used to enable real-time updates for ticket statuses, and the design was optimized for ease of use on both desktop and mobile devices. Testing was conducted both internally and with real users to identify and resolve issues, particularly related to usability and data accuracy. Challenges such as ensuring smooth real-time interactions and maintaining data consistency were addressed through careful validation and error handling. The system was successfully implemented in a local environment using XAMPP, which facilitated easy testing and development.

**Statistical Results Interpretation**

The evaluation of the SMFP Ticketing and Repair Management System was based on feedback collected from thirty-seven respondents consisting of thirty-two customers and five IT experts, using a Google Forms survey. The survey focused on assessing the system's usability, functionality, reliability, security, accessibility, compatibility and performance. The results were analyzed to understand how well the system met the needs of both customers and technicians, as well as its operational efficiency.

**User and IT Experts Rating**

The data from users and IT experts were collected based on their perspectives and choices, aligned with the ISO/IEC 25010 quality criteria.

***User Respondent***

| FUNCTIONALITY | Question | Weighted Mean | Standard Deviation | Interpretation |
|---|---|---|---|---|
| **Functional completeness** | Does the system provide an easy and convenient way for users to submit repair requests without difficulty? | 4.74 | 0.64 | **Strongly Agree** |

| | | | | |
|---|---|---|---|---|
| | | | | |
| **Functional correctness** | Does the system adequately meet the operational needs of SMFP Computer? | 4.83 | 0.70 | **Strongly Agree** |
| **Functional appropriateness** | Does the system properly manage repair jobs and service requests in an organized and efficient manner? | 3.67 | 0.68 | **Agree** |
| **General Weighted Mean Average** | | **4.41** | | **Agre** |

As gathered from the table above it shows the users' evaluation of the system's functionality using weighted means and standard deviations. Users generally agreed that the system allows easy submission of repair requests (4.74) and meets the operational needs of SMFP (4.83). They also agreed that the system manages repair jobs and service requests appropriately (3.57). The overall weighted mean of 4.41 indicates that, on average, respondents found the system's functionality acceptable.

*User Respondent*

| RELIABILITY | Question | Weighted Mean | Standard Deviation | Interpretation |
|---|---|---|---|---|
| **Availability** | Does the system provide consistent availability without unexpected downtime or interruptions? | 4.03 | 0.79 | **Agree** |
| **Fault tolerance** | Does the system maintain accurate and reliable records of tickets, repair jobs, and customer information? | 3.58 | 0.83 | **Agree** |
| **Recoverability** | Does the system consistently perform its intended functions without unexpected failures or issues? | 3.96 | 0.74 | **Agree** |

| | | | | |
|---|---|---|---|---|
| | | | | |
| **General Weighted Mean Average** | | **3.85** | | **Agree** |

The table shows user responses about the system's reliability. Users agreed that the system is generally available (4.03) and can continue to work through issues (3.96). They also agreed that it maintains accurate records (3.58), though this scored lower than the other items. The overall weighted mean of 3.85 indicates that users find the system's reliability acceptable.

***User Respondents***

| INTERACTION CAPABILITY | Question | Weighted Mean | Standard Deviation | Interpretation |
|---|---|---|---|---|
| **Appropriateness recognizability** | Does the system provide an intuitive interface and guidance that allows first-time users to perform tasks effectively? | 4.09 | 0.65 | **Agree** |

| | | | | |
|---|---|---|---|---|
| **Learnability** | Does the system allow users to quickly understand its functions and perform tasks without difficulty? | 3.80 | 0.83 | **Agree** |
| **Operability** | Does the system respond quickly and consistently to user actions such as submitting requests, viewing updates, or navigating pages? | 4.19 | 0.69 | **Agree** |
| **User error protection** | Does the system provide secure login and account management features that prevent unauthorized access? | 3.94 | 1.03 | **Agree** |
| **User interface aesthetics** | Does the system provide clear instructions and labels that help users | 4.09 | 0.79 | **Agree** |

| | | | | |
|---|---|---|---|---|
| | complete tasks without confusion? | | | |
| **Accessibility** | Does the system maintain full functionality and display correctly when used on different web browsers? | 3.93 | 0.83 | **Agree** |
| **Overall Experience** | Does the system meet users' expectations in supporting tasks and providing a smooth and reliable experience? | 3.93 | 0.85 | **Agree** |
| **General Weighted Mean Average** | | **4.05** | | **Agree** |

The table shows that users generally have a positive experience with the system. Most features, such as ease of use, responsiveness, error prevention, and interface design, received "Agree" ratings, with weighted means mostly above 4.00. This indicates that users find the system functional, understandable, and reliable. The general weighted mean of 4.05 suggests that, overall, the system meets user expectations and supports tasks effectively, though there is still room for minor improvement in areas like learnability and accessibility.

*IT Experts Respondents*

| FUNCTIONALITY | Question | Weighted Mean | Standard Deviation | Interpretation |
|---|---|---|---|---|
| **Functional completeness** | Does the system provide an easy and convenient way for users to submit repair requests without difficulty? | 3.4 | 0.55 | **Neutral** |
| **Functional correctness** | Does the system adequately meet the operational needs of SMFP Computer? | 3.0 | 0 | **Neutral** |
| **Functional appropriateness** | Does the system properly manage repair jobs and service requests in an organized and efficient manner? | 3.8 | 1.47 | **Agree** |

| | | | |
|---|---|---|---|
| **General Weighted Mean Average** | **3.4** | | **Neutral** |

The table shows how IT experts evaluated the system's functionality. The weighted means indicate that IT experts generally found the system neutral in terms of submitting repair requests and meeting operational needs, with values of 3.4 and 3.0, respectively. Functional appropriateness received a higher rating of 3.8, showing that IT experts agree the system manages repair jobs and service requests effectively. The standard deviations indicate variation in responses, with 0.55 and 0 showing consistent opinions for the first two questions, while 1.47 for functional appropriateness shows more varied experiences. Overall, the general weighted mean of 3.4 suggests that users see the system as adequately functional but with room for improvement.

***IT Experts Respondents***

| RELIABILITY | Question | Weighted Mean | Standard Deviation | Interpretation |
|---|---|---|---|---|
| **Availability** | Does the system provide consistent availability without unexpected downtime or interruptions? | 3.2 | 0.75 | **Neutral** |

| | | | | |
|---|---|---|---|---|
| **Fault tolerance** | Does the system maintain accurate and reliable records of tickets, repair jobs, and customer information? | 4.0 | 0.89 | **Agree** |
| **Recoverability** | Does the system consistently perform its intended functions without unexpected failures or issues? | 3.8 | 0.40 | **Agree** |
| **General Weighted Mean Average** | | **3.67** | | **Agree** |

The table shows IT experts' evaluation of the system's reliability. The weighted means indicate that IT experts found the system neutral in terms of availability (3.2) but agree that it is fault-tolerant (4.0) and recoverable (3.8). Standard deviations show some variation in responses, with 0.75 for availability, 0.89 for fault tolerance, and 0.40 for recoverability. Overall, the general weighted mean of 3.67 indicates that IT experts consider the system to be generally reliable, performing its intended functions accurately and consistently.

*IT Experts Respondents*

| INTERACTION CAPABILITY | Question | Weighted Mean | Standard Deviation | Interpretation |
|---|---|---|---|---|
| **Appropriateness recognizability** | Does the system provide an intuitive interface and guidance that allows first-time users to perform tasks effectively? | 3.40 | 1.02 | **Neutral** |
| **Learnability** | Does the system allow users to quickly understand its functions and perform tasks without difficulty? | 3.80 | 0.75 | **Agree** |
| **Operability** | Does the system respond quickly and consistently to user | 3.6 | 0.80 | **Agree** |

| | | | | |
|---|---|---|---|---|
| | actions such as submitting requests, viewing updates, or navigating pages? | | | |
| **User error protection** | Does the system provide secure login and account management features that prevent unauthorized access? | 3.4 | 0.80 | **Neutral** |
| **User interface aesthetics** | Does the system provide clear instructions and labels that help users complete tasks without confusion? | 3.6 | 0.80 | **Agree** |
| **Accessibility** | Does the system maintain full functionality and display correctly | 4.0 | 0.63 | **Agree** |

| | | | | |
|---|---|---|---|---|
| | when used on different web browsers? | | | |
| **Overall Experience** | Does the system meet users' expectations in supporting tasks and providing a smooth and reliable experience? | 4.0 | 0.63 | **Agree** |
| **General Weighted Mean Average** | | **3.68** | | **Agree** |

The table shows IT experts' evaluation of the system's interaction capability. The weighted means indicate that they rated the system neutral for appropriateness recognizability (3.40) and user error protection (3.4), but agree that it performs well in learnability (3.80), operability (3.6), user interface aesthetics (3.6), accessibility (4.0), and overall experience (4.0). Standard deviations range from 0.63 to 1.02, showing some variation in opinions, with the highest variation in appropriateness recognizability. Overall, the general weighted mean of 3.68 suggests that IT experts consider the system's interaction capability to be generally satisfactory and effective for supporting tasks.

# CHAPTER IV

## Summary of Findings, Conclusion, and Recommendation

This chapter presents a comprehensive summary of the findings derived from the system evaluation, followed by detailed conclusions and practical recommendations. The discussions in this chapter are based on the statistical results and interpretations presented in Chapter III, which assessed the effectiveness, usability, reliability, security, accessibility, and overall performance of the developed repair and service management system for SMFP Computer.

## Summary of Findings

The primary objective of this study was to design, develop, and evaluate a repair and service management system that would enhance customer service operations at SMFP Computer. To determine the effectiveness of the system, a structured evaluation was conducted involving 31 respondents using a five-point Likert scale questionnaire. The system was assessed across seven key quality dimensions.

Findings revealed that the system effectively fulfills its functional requirements. Most respondents expressed strong agreement that the system enables easy submission of repair requests and supports efficient management of service jobs. This indicates that the system successfully addresses the primary operational challenges previously encountered by SMFP Computer, such as manual tracking, delayed communication, and inefficient service workflows.

In terms of performance and responsiveness, the system received favorable ratings. Respondents agreed that the system performs reliably and responds promptly to user actions. Timely email notifications were also highlighted as an important feature that improved communication between customers and service personnel. These results demonstrate that the system supports faster service processing and enhances customer engagement.

The evaluation of usability and user experience showed that the system is generally easy to learn and use. Respondents agreed that system instructions, labels, and navigation are clear and understandable. The system's user-friendly interface allows first-time users to interact with minimal difficulty, reducing the need for extensive training. However, some neutral responses indicate that certain users may still experience minor difficulties, suggesting potential areas for interface refinement.

Regarding reliability, respondents agreed that the system is consistently available and operates without frequent errors. The accurate saving of repair tickets and service records further confirms the system's reliability and data integrity. These findings are critical, as reliable system availability and accurate data storage are essential for maintaining trust and ensuring uninterrupted service operations.

The findings related to security and data protection indicate that users generally trust the system when submitting service requests and managing their accounts. Respondents agreed that login and account security measures are effective and that the system prevents unauthorized changes to records. Although some users expressed neutral opinions, overall results suggest that the system provides adequate protection of sensitive data.

The system also demonstrated strong accessibility and compatibility. Respondents agreed that the system functions well across different web browsers and adapts effectively to various screen sizes. This confirms that the system supports convenient online access using different devices, which is essential in today's technology-driven environment.

Finally, overall satisfaction results indicate a high level of user acceptance. Respondents agreed that the system improves customer service efficiency and enhances communication between customers and SMFP staff. Most users expressed satisfaction with the system's overall performance, indicating that the system meets user expectations and delivers practical benefits to both customers and the organization.

**Recommendations**

Based on the comprehensive findings of the study, the following recommendations are proposed to further enhance the system and maximize its effectiveness:

**System Performance Enhancement**

Continuous performance monitoring and optimization should be conducted to ensure consistent system responsiveness, particularly during periods of high user activity.

**Usability Improvements**

Additional usability features such as interactive tooltips, guided tutorials, and contextual help menus should be incorporated to further improve ease of use and reduce user errors.

### Strengthening Security Measures

While current security features are adequate, additional mechanisms such as password strength indicators, multi-factor authentication, and login alerts could further enhance system security and user confidence.

### Improved Mobile Responsiveness

Further optimization for mobile devices is recommended to ensure seamless system access on smartphones and tablets, especially for users who rely on mobile platforms.

### Regular Maintenance and Updates

Scheduled system maintenance and updates should be implemented to address bugs, improve performance, and adapt to evolving user needs.

### User Training and Documentation

Providing user manuals, video tutorials, or orientation sessions for both staff and customers can help maximize system utilization and minimize operational errors.

### Future System Enhancements

Future development may include advanced features such as service analytics dashboards, automated reports, SMS notifications, and integration with inventory management systems.

### Conclusions

Based on the results of the system evaluation, it is concluded that the developed repair and service management system for SMFP Computer is effective, reliable, secure, and user-friendly. The system successfully meets its intended objectives by improving service request handling, enhancing communication, and streamlining customer service operations.

The high level of agreement across all evaluation criteria reflects strong user acceptance and satisfaction. The system's ability to improve efficiency and service quality demonstrates its value as a technological solution to existing operational challenges.

Although minor issues were identified in areas such as usability clarity and security visibility, these do not significantly affect the system's overall performance. With continuous improvement and proper system management, the system has strong potential for long-term use and scalability.

In conclusion, the developed system is recommended for full implementation at SMFP Computer and may serve as a foundation for future system enhancements. The study confirms that adopting a web-based repair and service management system can significantly improve customer service delivery and organizational efficiency.

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**APPENDICES**

**Appendix A: Letter Request for Client**

**NATIONAL TEACHERS COLLEGE**

March 14, 2025

Sherlo Pilarca
*Owner*
*SMFP Computer*
594 Nepomuceno St,
Quiapo, Manila, 1011

Good Day!

National Teachers College is the country's recognized leader and pioneer in High-tech education since 1928, continuing its mission to offer the most in-demand courses wherein graduates are most in-demand today. To keep pace with the rapid advancement of technology, without sacrificing thorough training, the students must undergo system development writing.

National Teachers College nurtures its students to develop to their full potential. Completely supported by the school through its faculty and staff, the delivery of the total academic, co-curricular and extracurricular activities make our students outshine all others by being proficient in their chosen fields and as fully integrated professionals capable of adapting to different real world scenarios.

In lieu of this, we would like to request your good office/institution to allow our third year students under the program of **Bachelor of Science in Information Technology** to conduct their studies in your company for compliance and completion in their Capstone Project.

Should there be any inquiries regarding this matter, Please feel free to call the undersigned at these mobile numbers 09617083480.

Thank you for your kind consideration.

**Proponents:**
Buhanghang, Jay
Diamante, James
Madera, Adrian
Quiñones, Michael
Yu, John Wilson

Yours truly

**Darwin C. Vargas**
*Capstone I Instructor*

**SMFP Computer Solutions**
*Company (Client)*

**Letter Request for Respondents**

**LETTER OF REQUEST TO THE RESPONDENTS**

**National Teachers College**

**School of Arts Sciences, and Technology**

**Dear Respondent:**

**Please be informed that we are conducting a capstone project entitled**

**SmartLink: A Unified Platform for IT Services and Customer Support in fulfillment of the degree, Bachelor of Science in Information Technology. In this connection, we have constructed a questionnaire to evaluate the acceptability of the developed software product. Your participation in the study by way of answering the questionnaire is very vital. Without it, the study will not be as complete as it should be. Please feel assured that your anonymity and the information you will give will be treated with the strictest confidentiality.**

**Thank you very much for your very kind response to our request and if you are interested, we will supply you with the results of our research project.**

**Very sincerely yours,**

## Appendix B: Evaluation Instrument (User and IT Expert)

The link below is the SMFP Demo video:

https://drive.google.com/file/d/1oYZO9RKO3Oxi6sZb2oHbF1aLNICIsUq2/view?usp=sharing

Instructions

Please read each statement carefully and select the response that best reflects your experience using the system.

**Rating Scale:**

5 - Strongly Agree
4 - Agree
3 - Neutral
2 - Disagree
1 - Strongly Disagree

Does the system adequately meet the operational needs of SMFP Computer?

- ○ 5 - Strongly Agree
- ○ 4 - Agree
- ○ 3 - Neutral
- ○ 2 - Disagree
- ○ 1 - Strongly Disagree

*Does the system properly manage repair jobs and service requests in an organized and efficient manner?*

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Part II. Performance and Responsiveness

*(Speed and efficiency of the system)*

Does the system send email notifications promptly to inform users about important updates in repair status?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system perform satisfactorily under normal usage conditions without frequent errors or delays?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system respond quickly and consistently to user actions such as submitting requests, viewing updates, or navigating pages?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Part III. Usability and User Experience

*(Ease of use and interface design)*

Does the system provide clear instructions and labels that help users complete tasks without confusion?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system provide an intuitive interface and guidance that allows first-time users to perform tasks effectively?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system allow users to quickly understand its functions and perform tasks without difficulty?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Part IV. Reliability

*(System stability and consistency)*

Does the system provide consistent availability without unexpected downtime or interruptions?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system consistently perform its intended functions without unexpected failures or issues?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system maintain accurate and reliable records of tickets, repair jobs, and customer information?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Part V. Security and Data Protection

*(Protection of user information and system access)*

Does the system provide enough guidance and clarity for users to feel confident when submitting service requests?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system protect records from unauthorized access or modifications?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system provide secure login and account management features that prevent unauthorized access?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Part VI. Accessibility and Compatibility

*(Device and platform support)*

Does the system maintain full functionality and display correctly when used on different web browsers?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system maintain usability and proper layout when accessed on screens of different sizes?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system enable users to access and use services online without difficulty or unnecessary steps?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Part VII. Overall Satisfaction

Description (optional)

Does the system help staff manage customer service tasks more efficiently, resulting in faster and more organized service delivery?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system help staff and customers communicate more effectively, reducing delays and misunderstandings in service management?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Does the system meet users' expectations in supporting tasks and providing a smooth and reliable experience?

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

## Appendix C: Summary of Evaluation Results

*(Number of Users - 32 and IT Experts - 5)*

| *SOFTWARE EVALUATION CHARACTERISTICS* | Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree |
|---|---|---|---|---|---|
| **FUNCTIONALITY** | | | | | |
| Functional completeness | 14 | 13 | 7 | 2 | 1 |
| Functional correctness | 14 | 13 | 8 | 1 | 1 |
| Functional appropriateness | 11 | 17 | 6 | 2 | 1 |
| **RELIABILITY** | | | | | |
| Availability | 13 | 11 | 11 | 1 | 1 |
| Fault tolerance | 11 | 16 | 7 | 1 | 2 |
| Recoverability | 10 | 17 | 9 | 0 | 1 |
| **INTERACTION CAPABILITY** | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| Appropriateness recognizability | 14 | 13 | 7 | 2 | 1 |
| Learnability | 12 | 16 | 3 | 2 | 4 |
| Operability | 12 | 19 | 5 | | 1 |
| User error protection | 10 | 17 | 7 | 1 | 2 |
| User interface aesthetics | 15 | 14 | 5 | | 3 |
| Accessibility | 10 | 20 | 5 | | 2 |
| Overall Experience of Users | 11 | 17 | 7 | | 2 |

**Summary of Evaluation Responses of Users**

| Criteria | Strongly Disagree (1) | Disagree (2) | Neutra 1 (3) | Agree (4) | Strongly Agree (5) | Interpr etation |
|---|---|---|---|---|---|---|
| **FUNCTION ALITY** | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Functional completeness** | 1 | 2 | 4 | 11 | 14 | **Strongly Agree** |
| **Functional correctness** | 1 | 1 | 3 | 13 | 14 | **Strongly Agree** |
| **Functional appropriateness** | 1 | 1 | 6 | 15 | 9 | **Agree** |
| **RELIABILITY** | | | | | | |
| **Availability** | 1 | | 9 | 9 | 13 | **Agree** |
| **Fault tolerance** | 2 | 1 | 5 | 15 | 9 | **Agree** |
| **Recoverability** | 1 | | 8 | 13 | 10 | **Agree** |
| **INTERACTION CAPABILITY** | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Appropriateness recognizability** | 1 | 1 | 5 | 12 | 13 | **Agree** |
| **Learnability** | 4 | 2 | 1 | 14 | 11 | **Agree** |
| **Operability** | 1 | | 2 | 18 | 11 | **Agree** |
| **User error protection** | 2 | | 6 | 14 | 10 | **Agree** |
| **User interface aesthetics** | 3 | | 2 | 13 | 14 | **Agree** |
| **Accessibility** | 2 | | 4 | 17 | 9 | **Agree** |
| **Overall Experience** | 2 | | | | 6 | **Agree** |

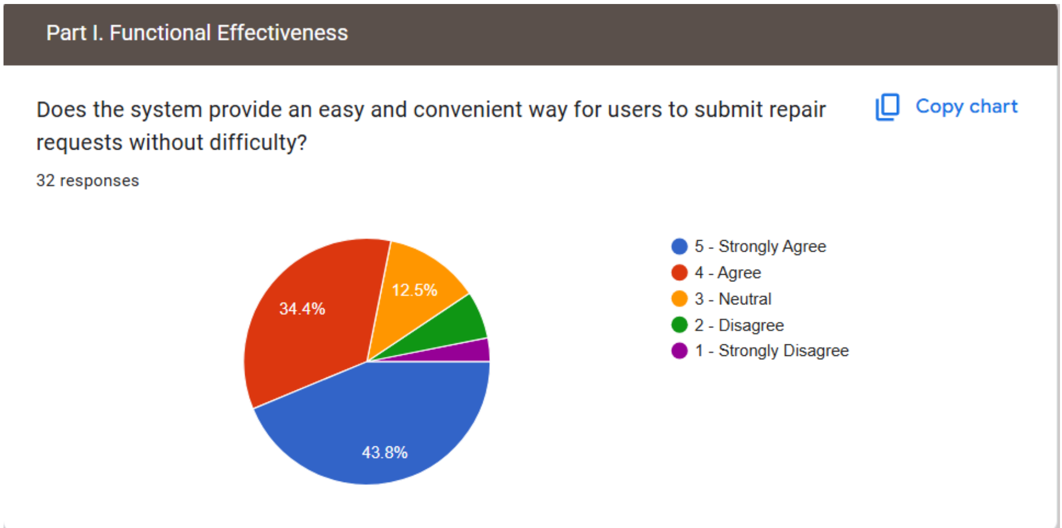
*Summary of Evaluation Responses of IT Experts*

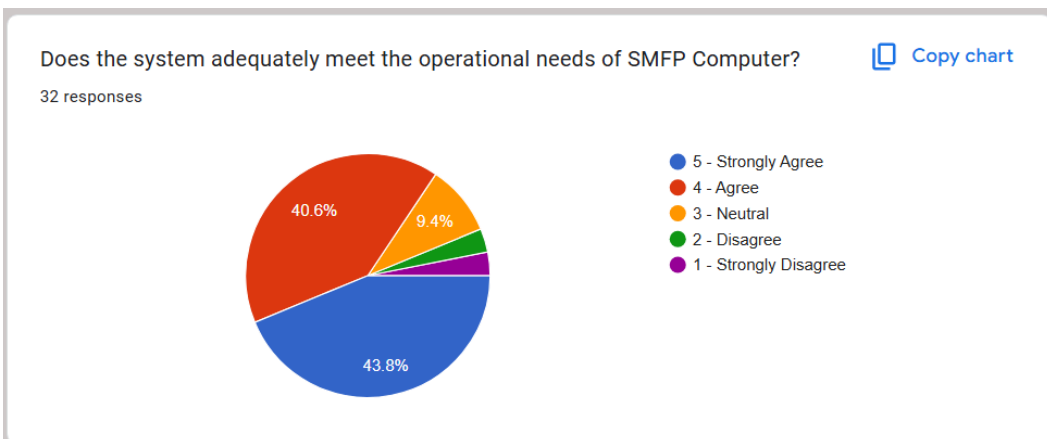
| Criteria | Strongly Disagree (1) | Disagree (2) | Neutral (3) | Agree (4) | Strongly Agree (5) | Interpretation |
|---|---|---|---|---|---|---|
| **FUNCTIONALITY** | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Functional completeness** | | | 3 | 2 | | **Neutral** |
| **Functional correctness** | | | 5 | | | **Neutral** |
| **Functional appropriateness** | | 1 | | 2 | 2 | **Agree** |
| **RELIABILITY** | | | | | | |
| **Availability** | | 1 | 2 | 2 | | **Neutral** |
| **Fault tolerance** | | | 2 | 1 | 2 | **Agree** |
| **Recoverability** | | | 1 | 4 | | **Agree** |
| **INTERACTION CAPABILITY** | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Appropriateness recognizability** | | 1 | 2 | 1 | 1 | **Neutral** |
| **Learnability** | | | 2 | 2 | 1 | **Agree** |
| **Operability** | | | 3 | 1 | 1 | **Agree** |
| **User error protection** | | 1 | 1 | 3 | | **Neutral** |
| **User interface aesthetics** | | | 3 | 1 | 1 | **Agree** |
| **Accessibility** | | | 1 | 3 | 1 | **Agree** |
| **Overall Experience** | | | 1 | 3 | 1 | **Agree** |

# Appendix D: Test Results

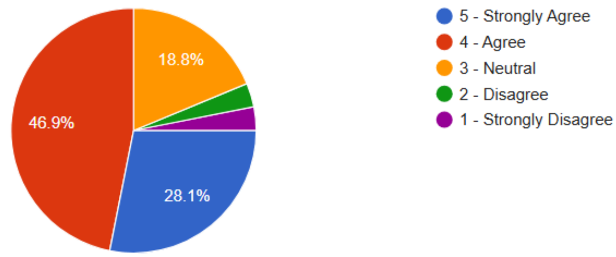


Copy chart

*Does the system properly manage repair jobs and service requests in an organized and efficient manner?*

32 responses

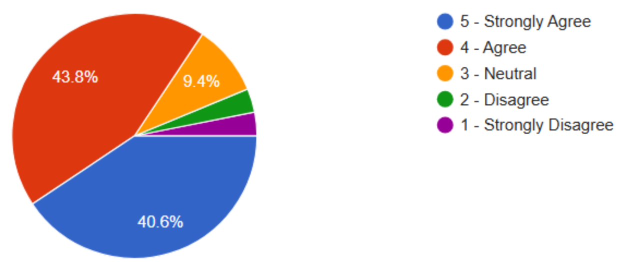



## Part II. Performance and Responsiveness

Copy chart

Does the system send email notifications promptly to inform users about important updates in repair status?

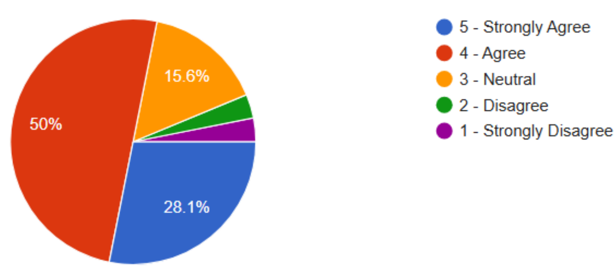
32 responses




Copy chart

Does the system perform satisfactorily under normal usage conditions without frequent errors or delays?

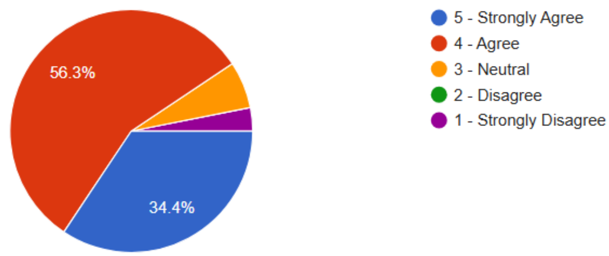
32 responses



Does the system respond quickly and consistently to user actions such as submitting requests, viewing updates, or navigating pages?

Copy chart

32 responses

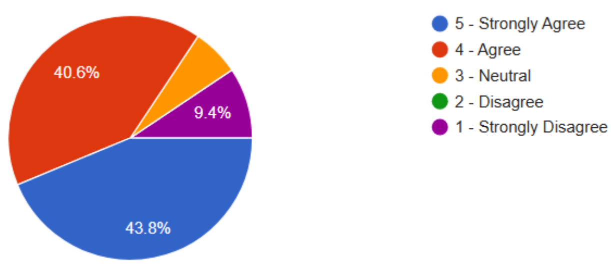



## Part III. Usability and User Experience

Does the system provide clear instructions and labels that help users complete tasks without confusion?

Copy chart

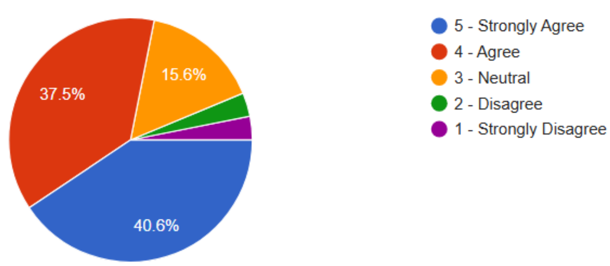
32 responses




Does the system provide an intuitive interface and guidance that allows first-time users to perform tasks effectively?

Copy chart

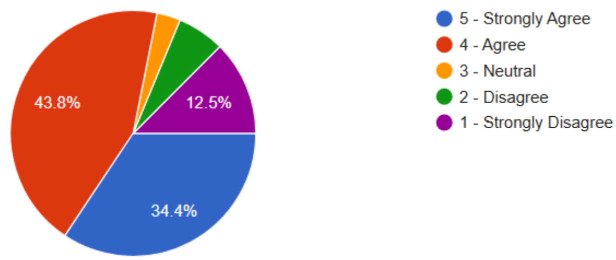
32 responses



Does the system allow users to quickly understand its functions and perform tasks without difficulty?

Copy chart

32 responses

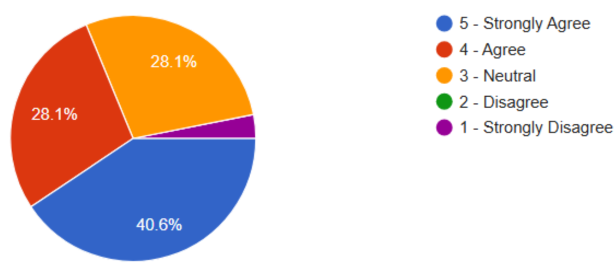



## Part IV. Reliability

Does the system provide consistent availability without unexpected downtime or interruptions?

Copy chart

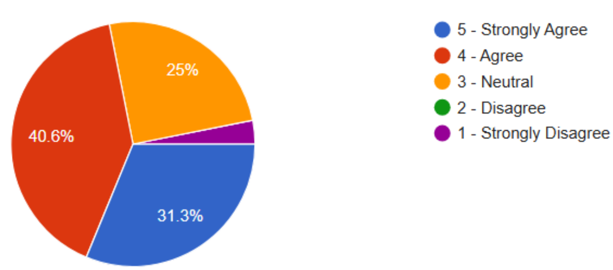
32 responses




Does the system consistently perform its intended functions without unexpected failures or issues?

Copy chart

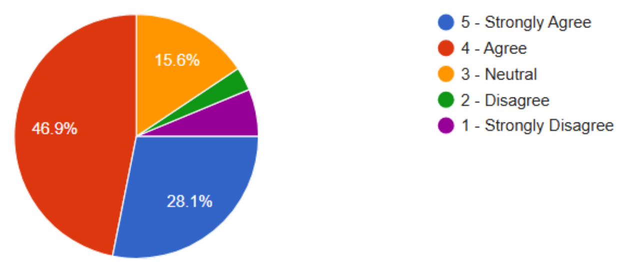
32 responses



Does the system maintain accurate and reliable records of tickets, repair jobs, and customer information?

Copy chart

32 responses

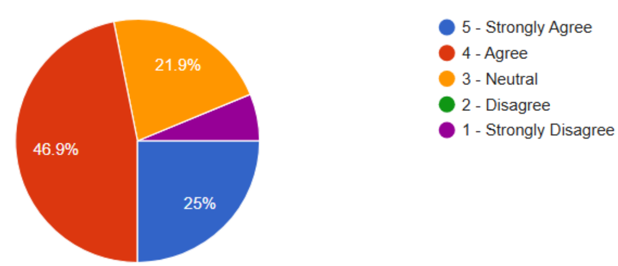



## Part V. Security and Data Protection

Does the system provide enough guidance and clarity for users to feel confident when submitting service requests?

Copy chart

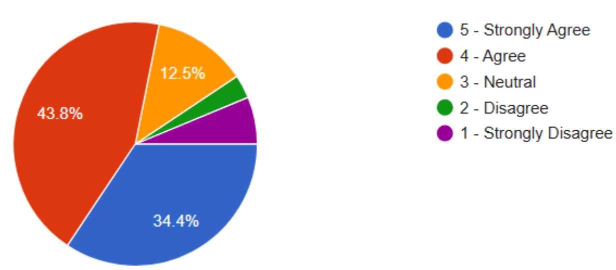
32 responses




Does the system protect records from unauthorized access or modifications?

Copy chart

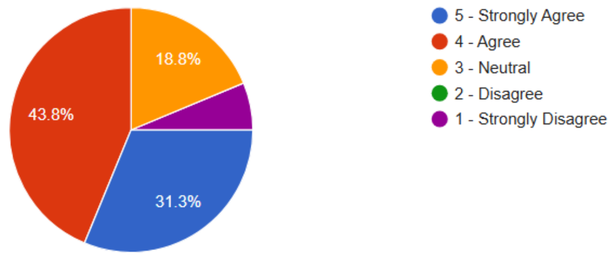
32 responses



Does the system provide secure login and account management features that prevent unauthorized access?

Copy chart

32 responses

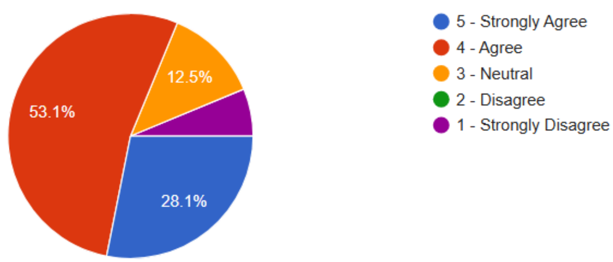



## Part VI. Accessibility and Compatibility

Does the system maintain full functionality and display correctly when used on different web browsers?

Copy chart

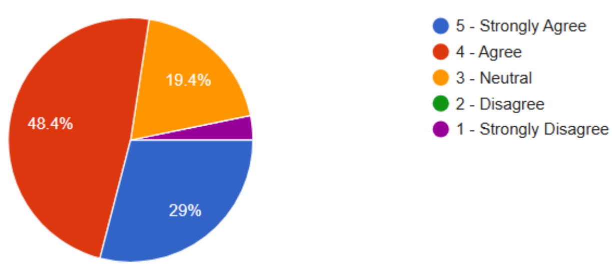
32 responses




Does the system maintain usability and proper layout when accessed on screens of different sizes?

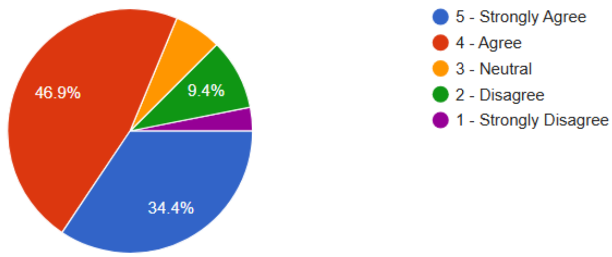
Copy chart

31 responses



Does the system enable users to access and use services online without difficulty or unnecessary steps?

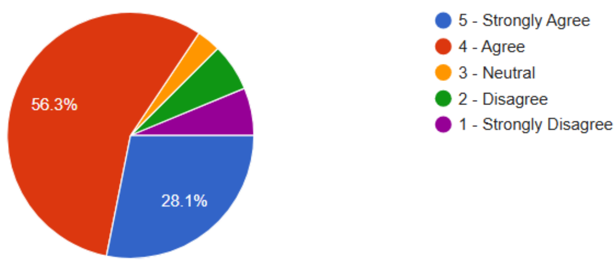
32 responses




## Part VII. Overall Satisfaction

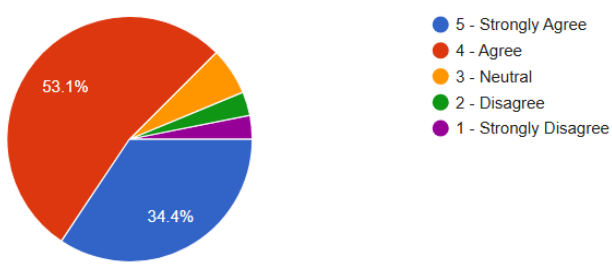
Does the system help staff manage customer service tasks more efficiently, resulting in faster and more organized service delivery?

32 responses




Does the system help staff and customers communicate more effectively, reducing delays and misunderstandings in service management?

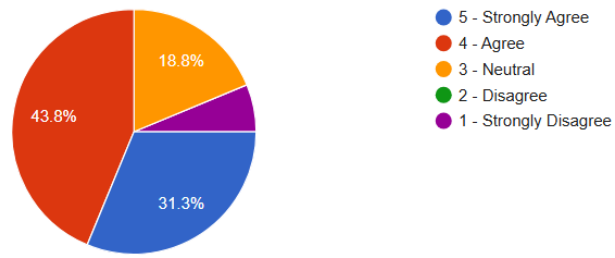
32 responses



Does the system meet users' expectations in supporting tasks and providing a smooth and reliable experience?

Copy chart

32 responses

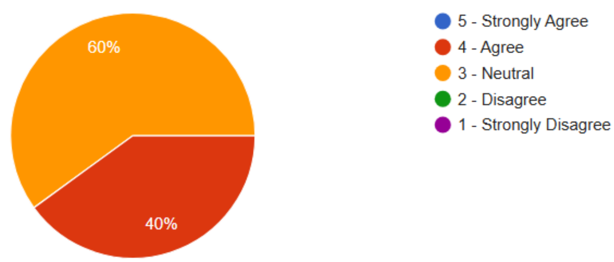



## FUNCTIONALITY

Copy chart

Does the system provide an easy and convenient way for users to submit repair requests without difficulty?

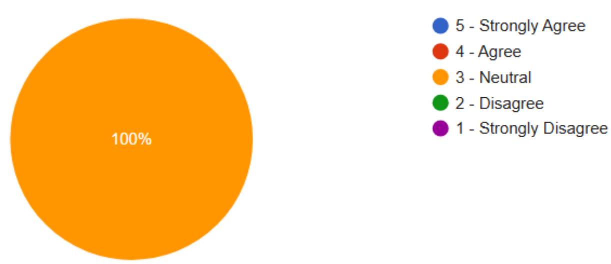
5 responses




Copy chart

Does the system adequately meet the operational needs of SMFP Computer?

5 responses



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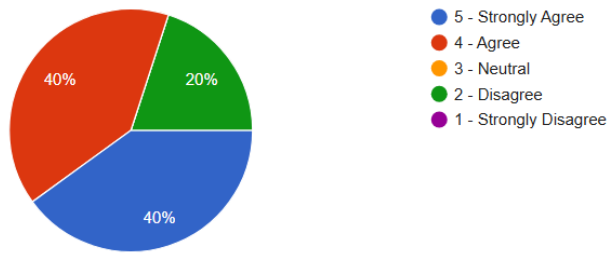
Does the system properly manage repair jobs and service requests in an organized and efficient manner?

5 responses

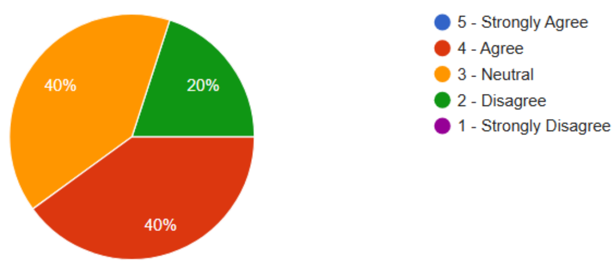



## RELIABILITY

Copy chart

Does the system provide consistent availability without unexpected downtime or interruptions?

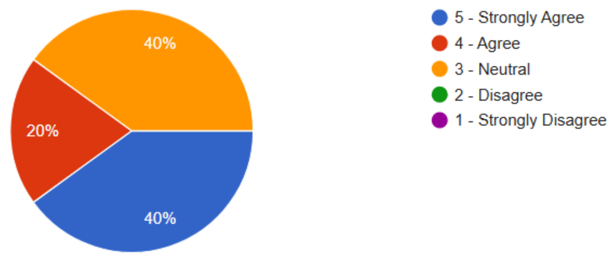
5 responses



Copy chart

Does the system maintain accurate and reliable records of tickets, repair jobs, and customer information?

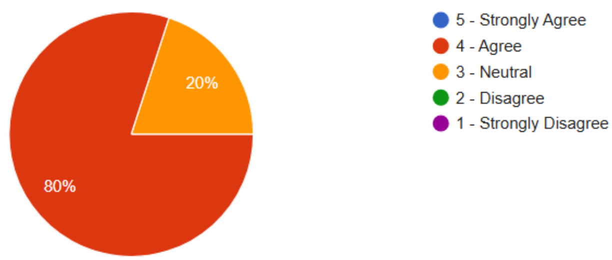
5 responses




Copy chart

Does the system consistently perform its intended functions without unexpected failures or issues?

5 responses

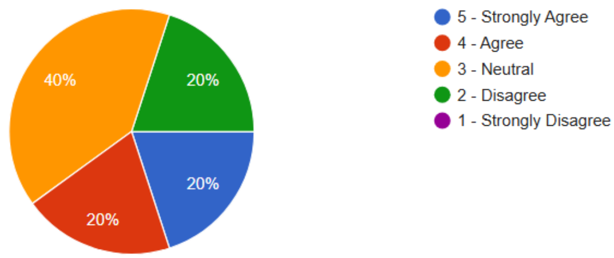


## INTERACTION CAPABILITY

Copy chart

Does the system provide an intuitive interface and guidance that allows first-time users to perform tasks effectively?

5 responses

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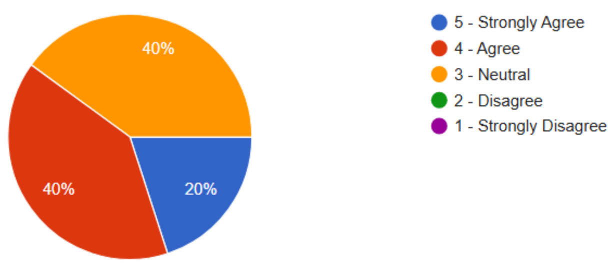
Does the system allow users to quickly understand its functions and perform tasks without difficulty?

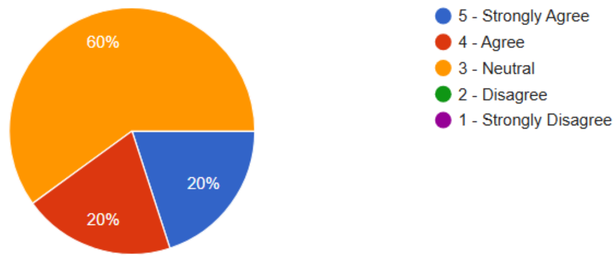
5 responses



Copy chart

Does the system respond quickly and consistently to user actions such as submitting requests, viewing updates, or navigating pages?

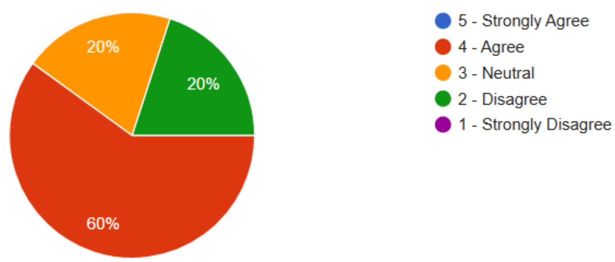
5 responses




Copy chart

Does the system help prevent mistakes (e.g., wrong booking input, repeated uploads)?

5 responses

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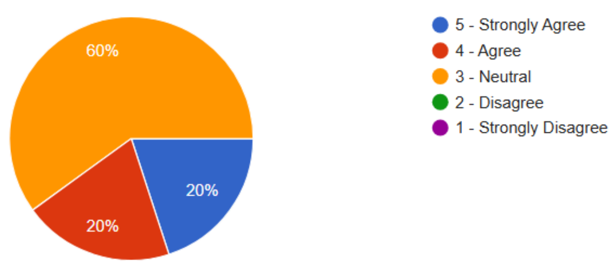
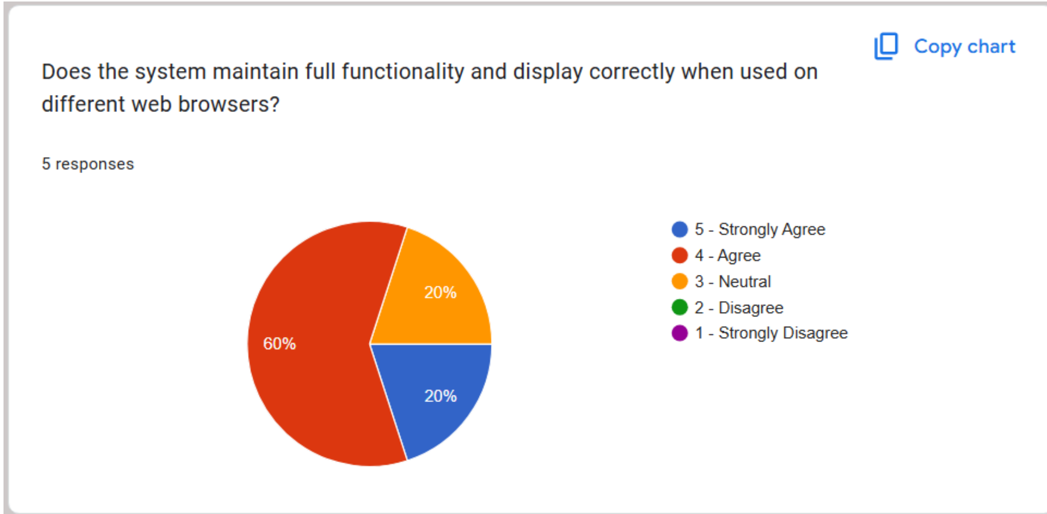
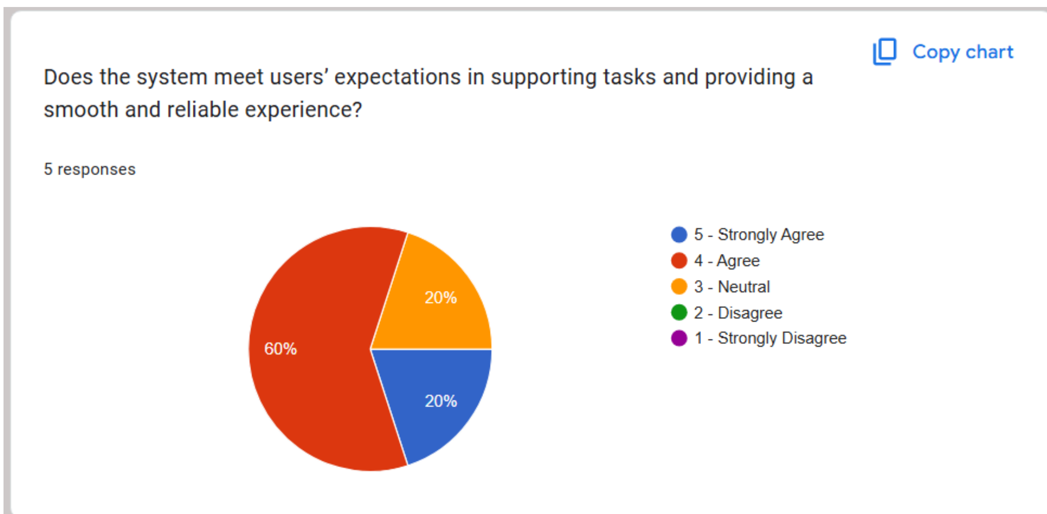
Does the system provide clear instructions and labels that help users complete tasks without confusion?

5 responses



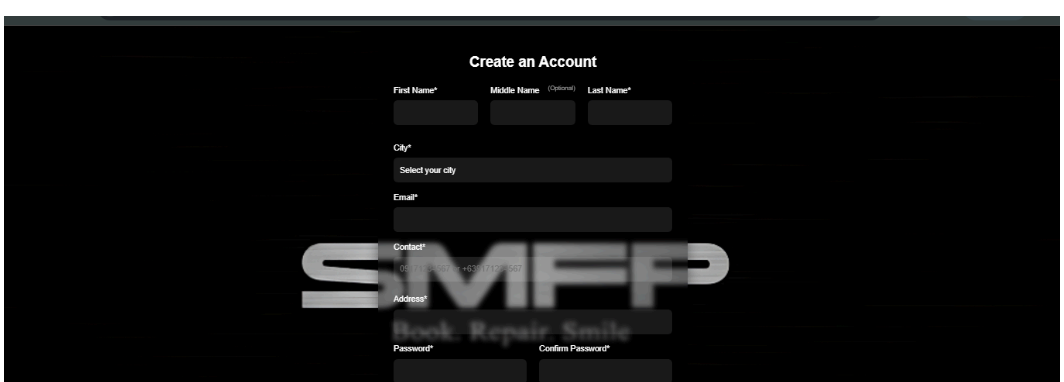


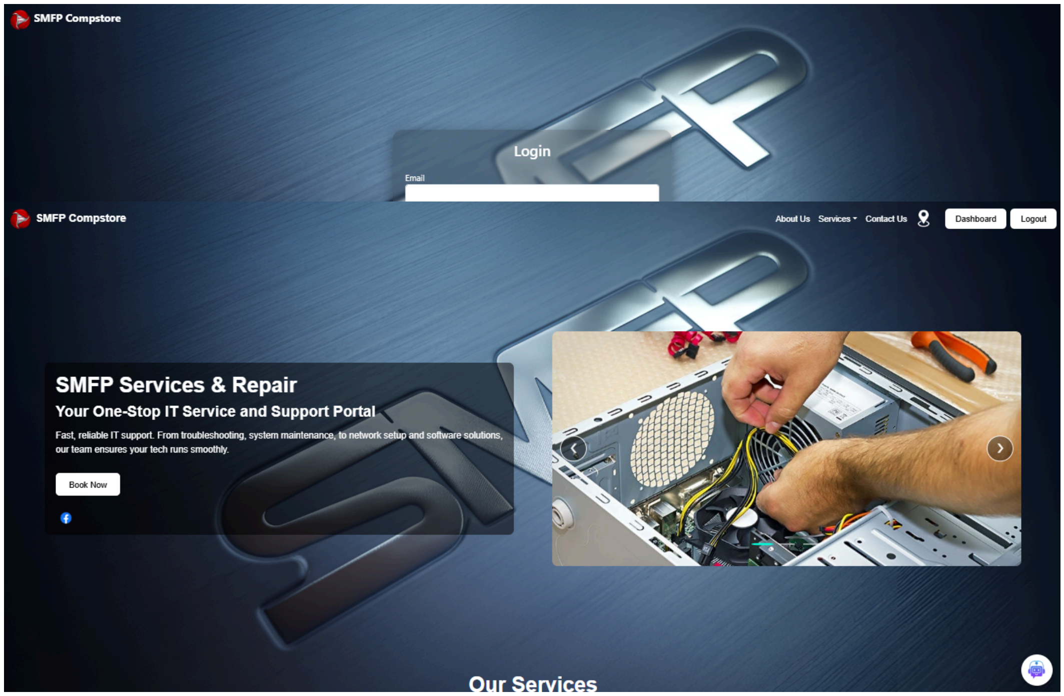

## Appendix E: System Screenshot

### USER PAGE

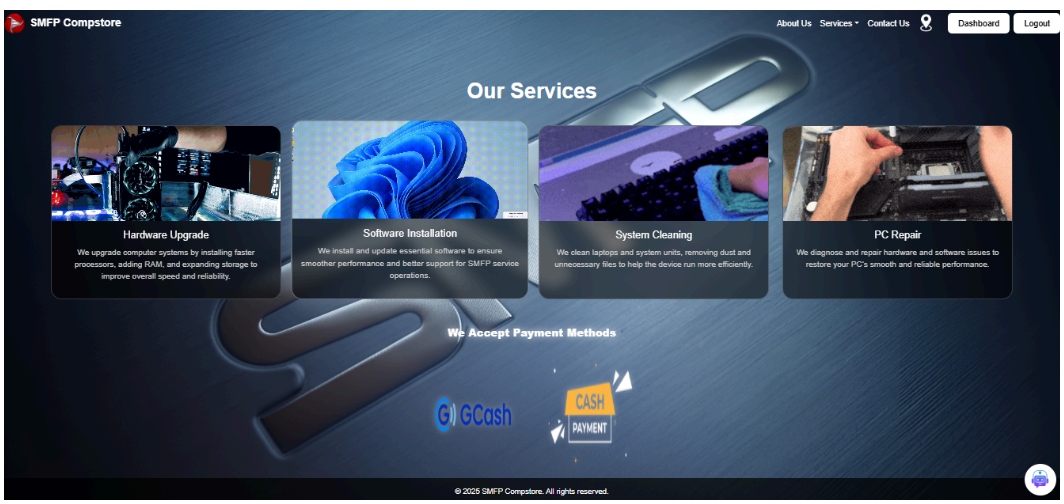
### Signup



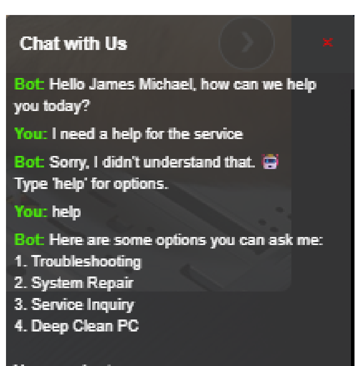
## Login



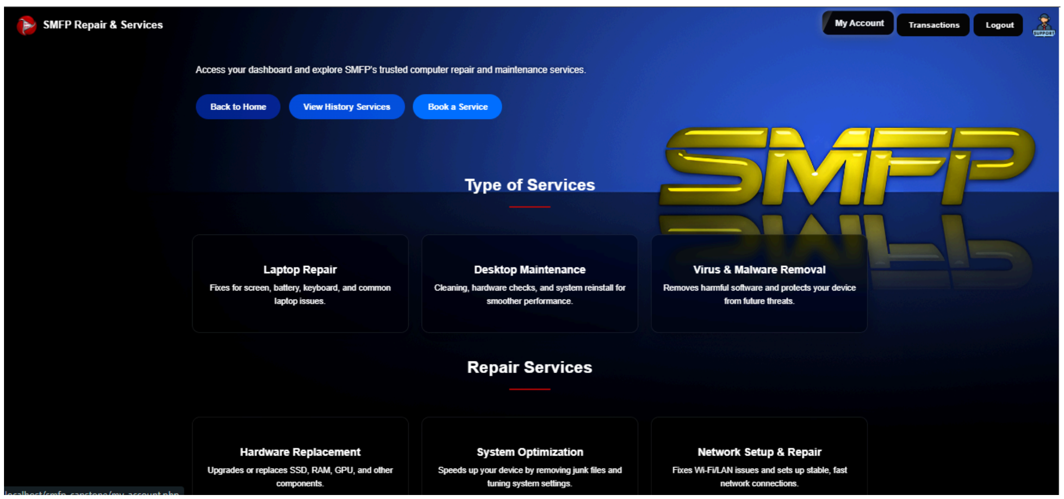

## Home Page




## Chatbot

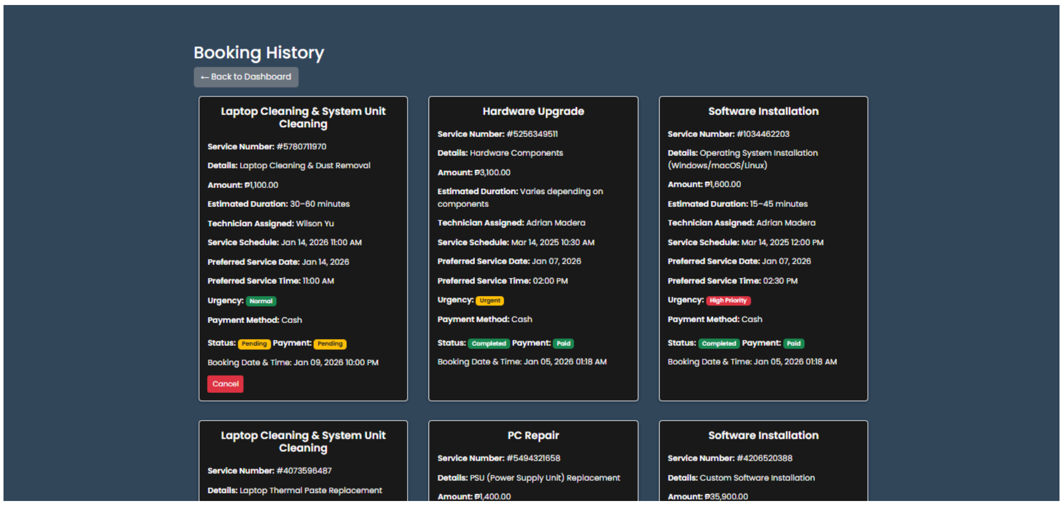


## Customer Dashboard

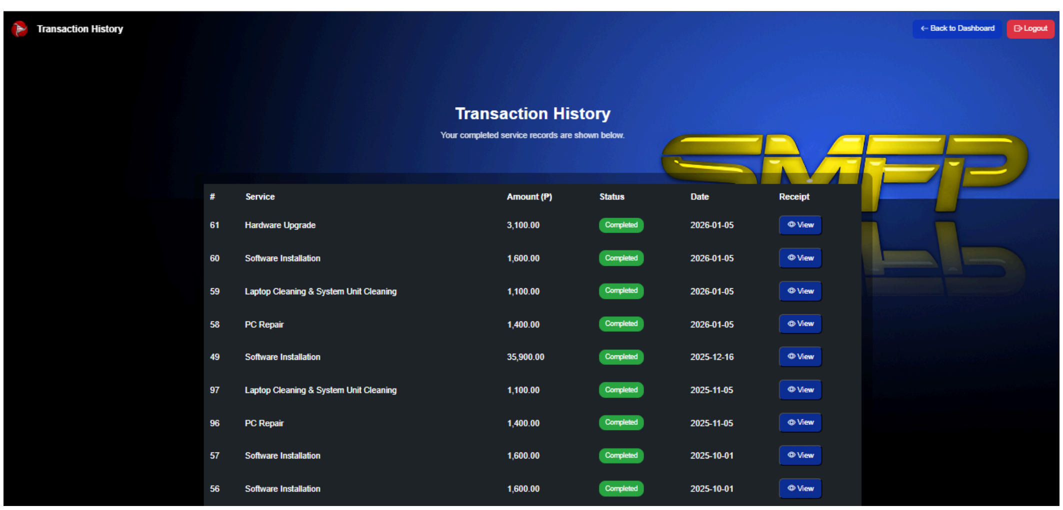



## Booking PC/Laptop Services

## Booking Summary & History




## Transaction History



## Receipt

**OFFICIAL RECEIPT**

**Business Details:** SMFP Computer, 594 J Nepomuceno St, Quiapo, Metro Manila
Phone: 0912 345 6789 | Email: smfpcompstore.support@gmail.com

**Customer Details:** Name: James Michael Diamante
Email: levi40694@gmail.com | Contact: 09082324554

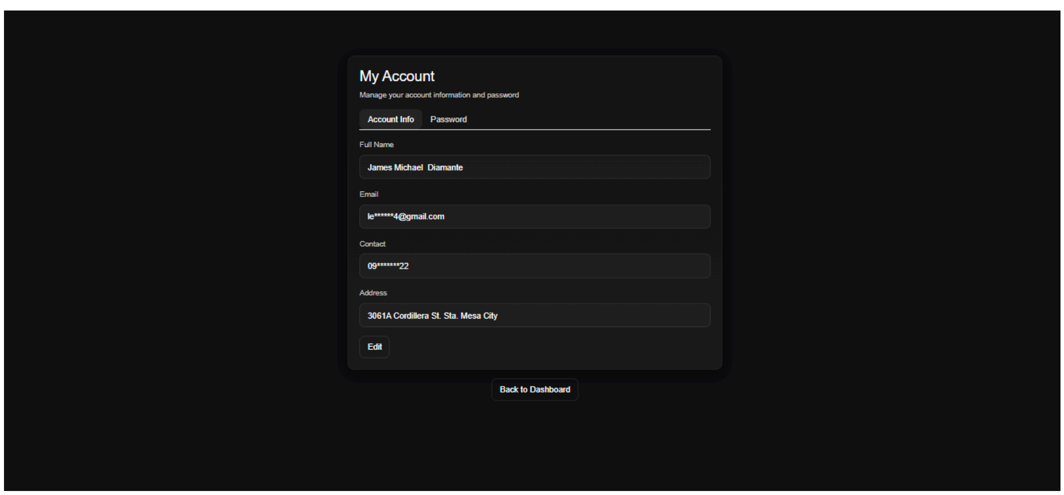
**Receipt Info:** Date: Jan 09, 2026
Service Number: #5780711970

**Itemized Services**

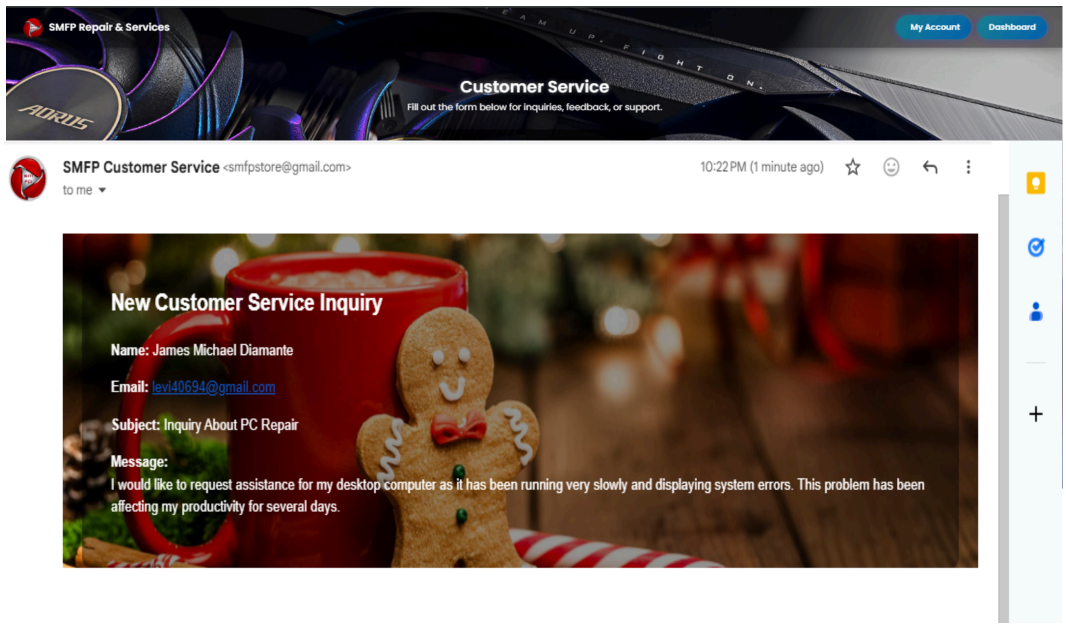
| Service/Item | Description | Unit Price | Total |
|---|---|---|---|
| Laptop Cleaning & System Unit C | Laptop Cleaning & Dust Removal | PHP. 500.00 | PHP. 500.00 |
| Components: N/A | N/A | PHP. 0.00 | PHP. 0.00 |
| Urgency Fee | Normal | PHP. 300.00 | PHP. 300.00 |

**Service Fee:** **PHP. 500.00**
**Labor Fee:** **PHP. 500.00**
**Component Fee:** **PHP. 0.00**
**Urgency Fee:** **PHP. 300.00**
**Total Amount:** **PHP. 1,300.00**

## Customer Account

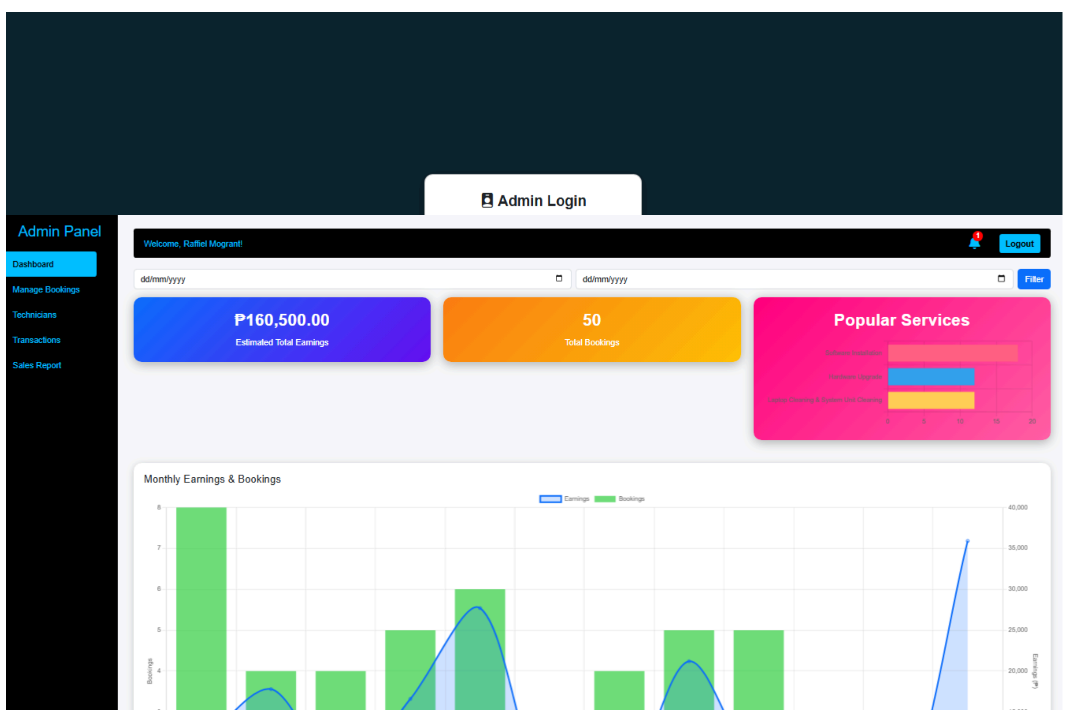


## Customer Service



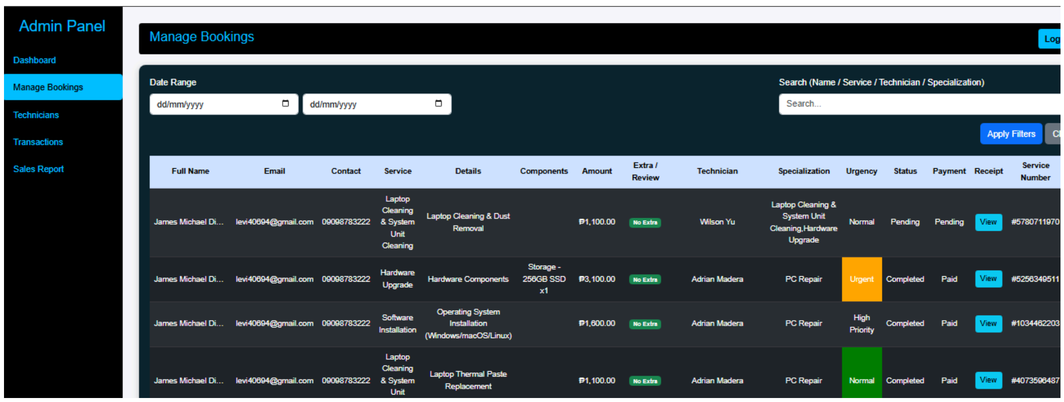

## Admin Page

## Login

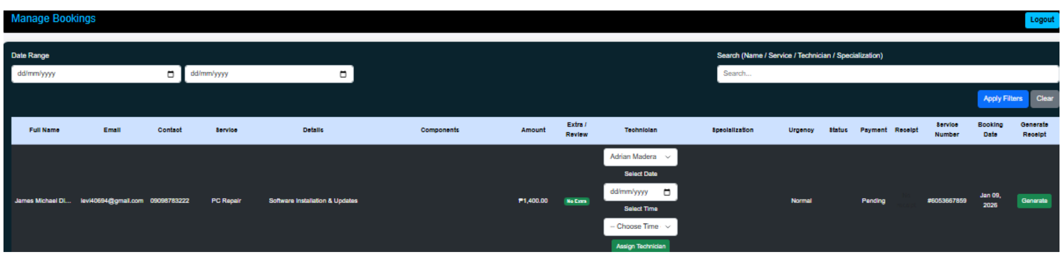



## Dashboard Overview

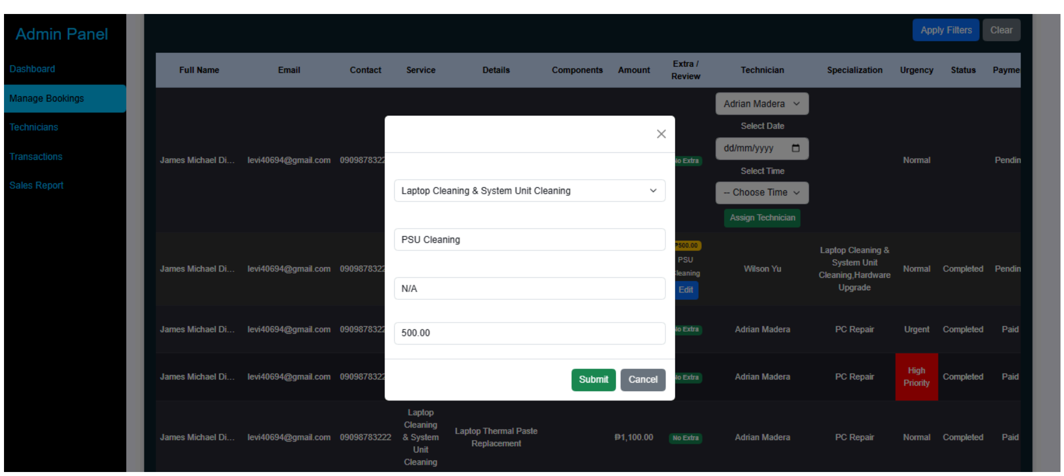
## Booking Management



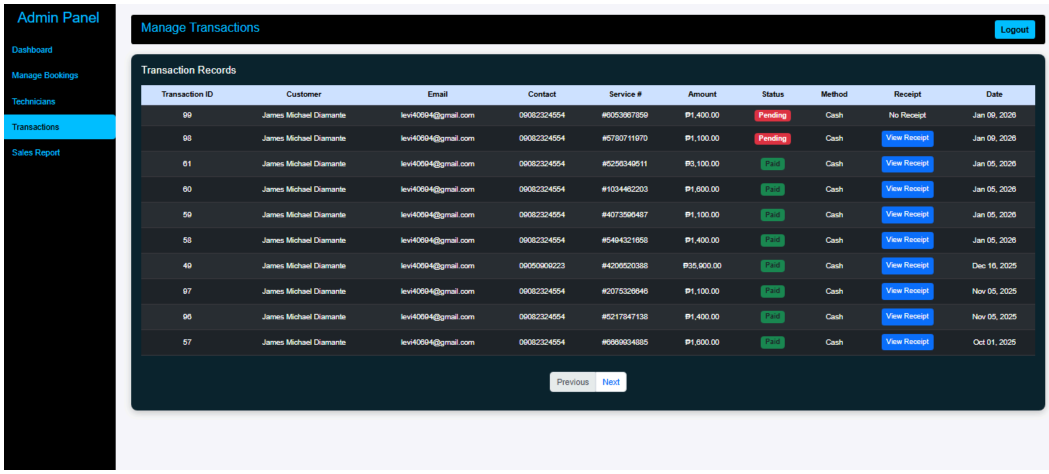

## Update Booking Technician



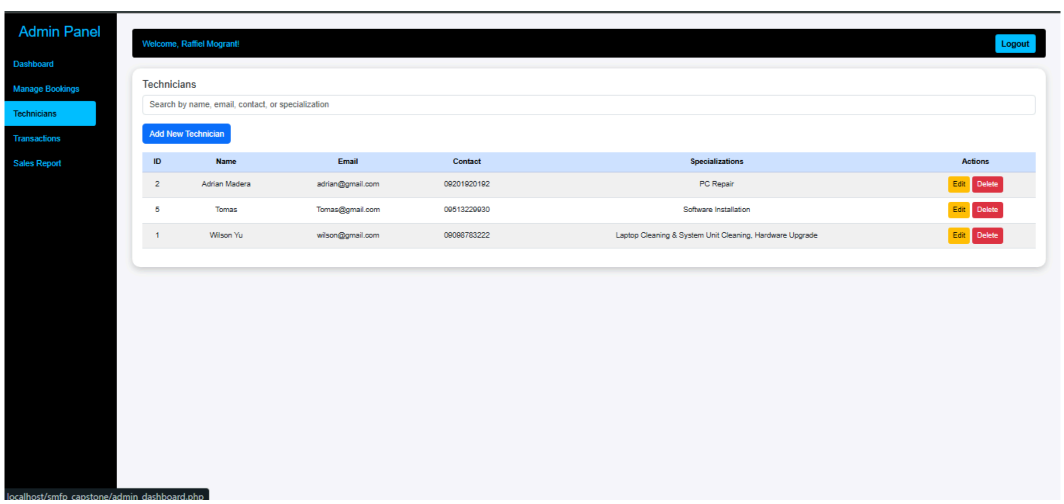

## Additional Service Confirmation



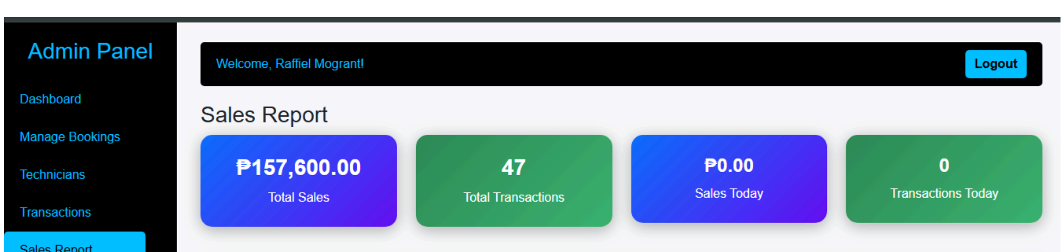
## Transaction Management

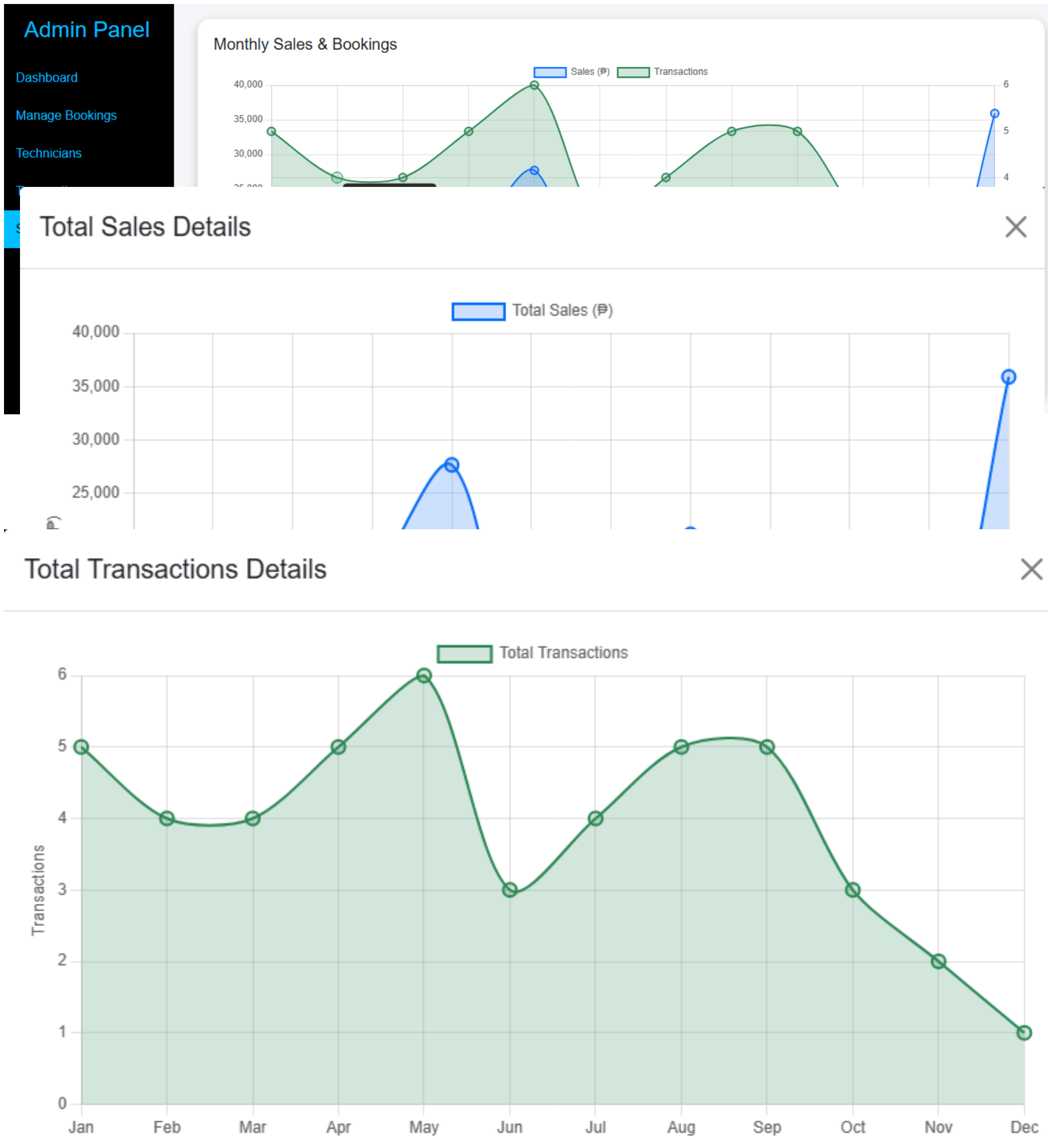



## Store Technicians



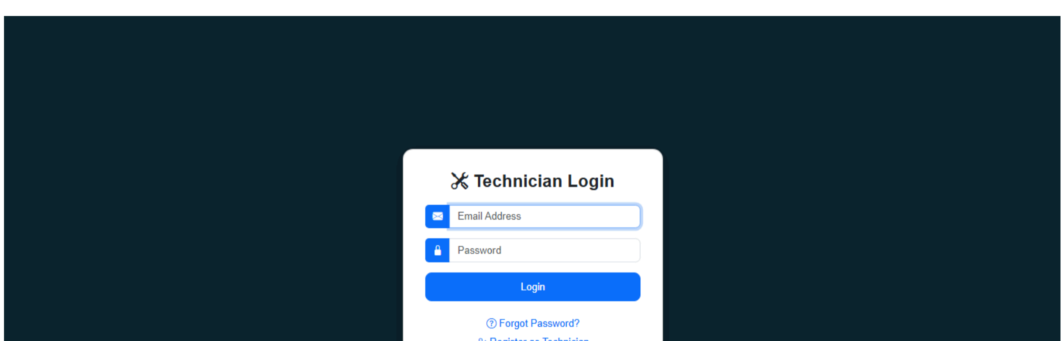

## Sales Reports



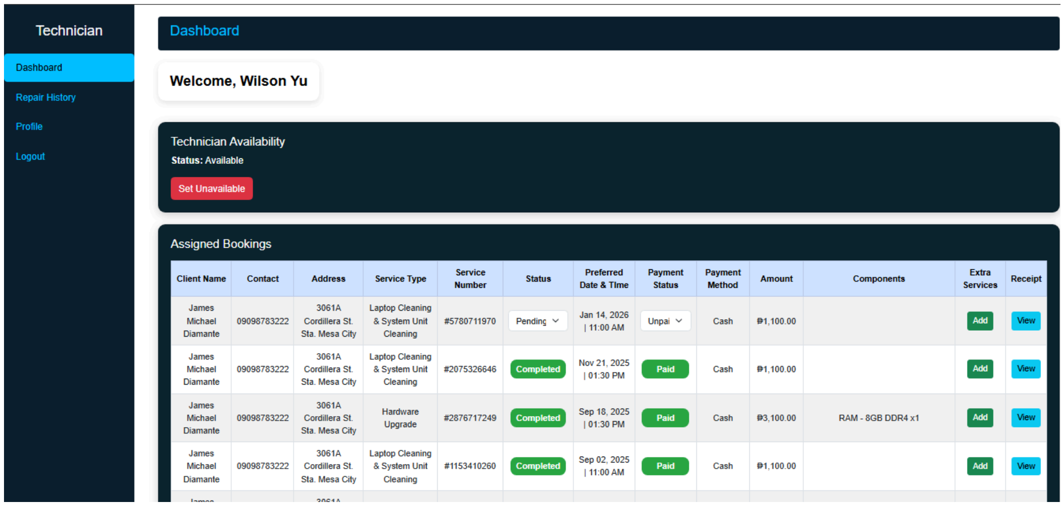



## Technician Page

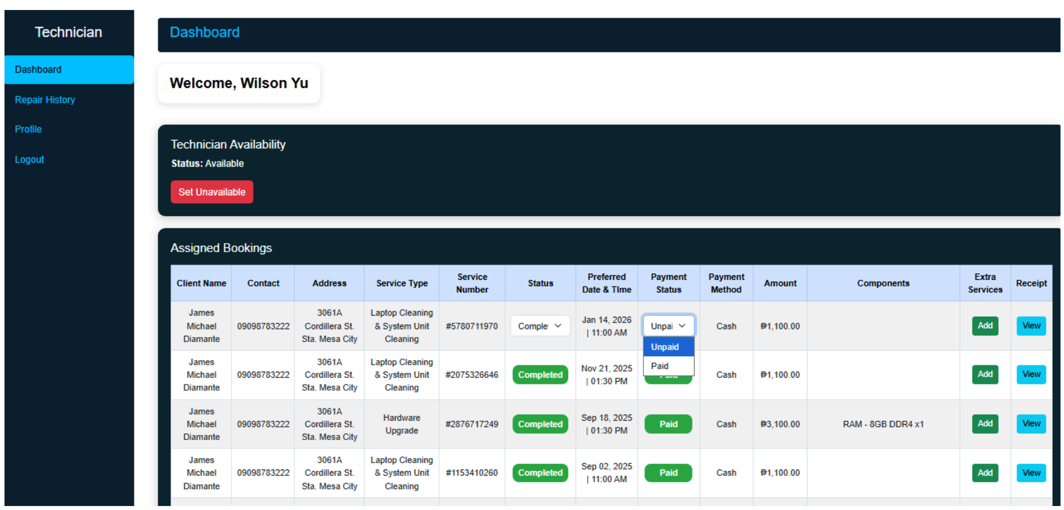
## Login



## Dashboard Overview



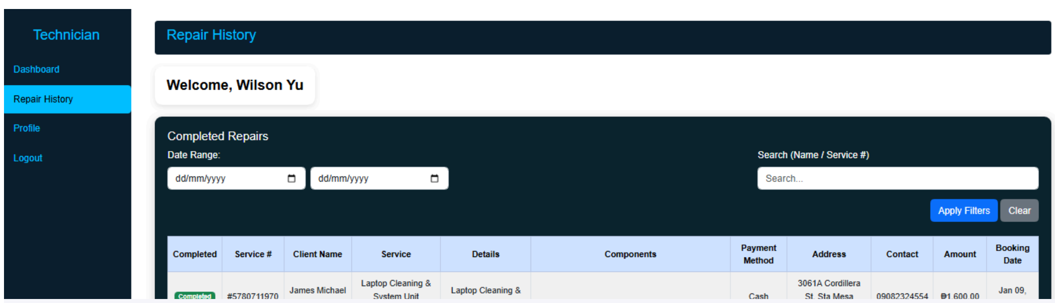

## Update Status & Payment Status

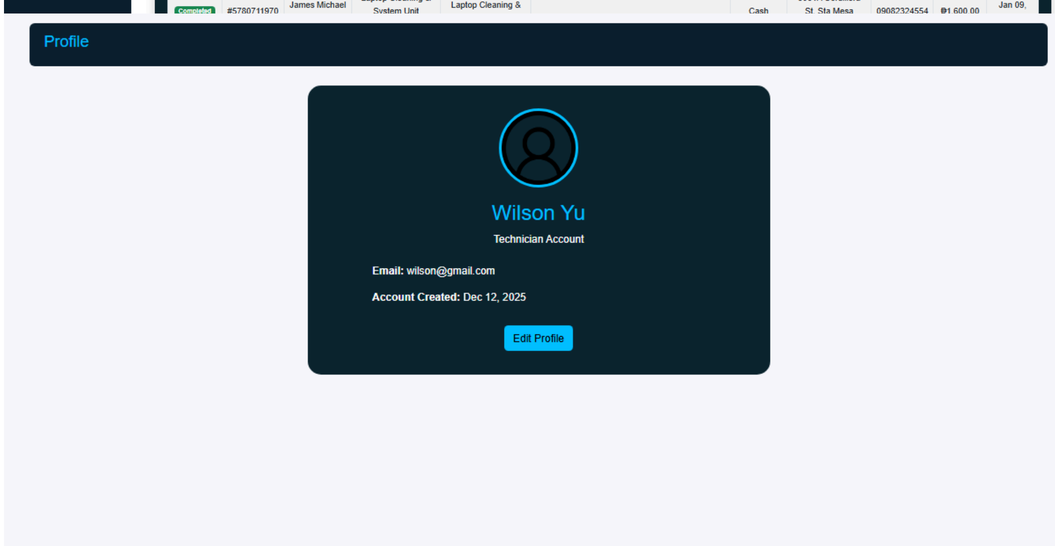



## Additional Services



## Repair History




## Technician Profile

# Appendix F: User Manual

## SMFP Repair & Services Website

Welcome to the SMFP Repair & Services Website. This guide is designed to help customers, technicians, and administrators navigate and use all features of the website efficiently and effectively.

## User Manual

This section provides step-by-step guidance for customers in submitting service requests, monitoring service progress, and managing their accounts.

## Requirements Before You Begin

- Device with a web browser (Google Chrome, Microsoft Edge, or Mozilla Firefox)
- Stable internet connection
- Valid email address for account registration

## Account Setup

### *Sign Up Process*

1. Open the system website.
2. Click Sign Up.
3. Fill in all required personal information.
4. Enter a valid email address and password.
5. Read and accept the Terms and Conditions.
6. Click Register to create your account.

### *Login*

1. Enter your registered email address and password.
2. Click Login to access the user dashboard.

## Getting Started

After logging in for the first time, the dashboard will be displayed, which includes:

- Service booking option
- Booking history
- Transactions page for completed bookings
- Account profile settings

### Booking

1. Click Book a Service from the dashboard.
2. Provide the required customer information.
3. Select the desired service type.
4. Enter additional details or special instructions if necessary.
5. Choose the preferred date and time.
6. Review the booking summary.
7. Click Submit Request to complete the booking.

Once submitted, the booking status will be set to Pending.

### Booking History and Tracking

The Booking History page allows users to monitor and manage their service requests:

- **Pending** – Booking has been submitted and is awaiting confirmation
- **In Progress** – Service work is currently ongoing
- **Completed** – Service has been successfully completed

- **Cancelled** – Booking has been cancelled by the user or administrator

Functions available in Booking History:

- View booking details and current status
- Cancel a booking if it is still pending or in progress

Note:

- When a booking is cancelled by the user, it is automatically removed from the booking history list.

To view booking history:

1. Navigate to Booking History from the user dashboard.
2. Select a booking to view its detailed information and status.

### Transactions

The Transactions page displays all completed service bookings:

- Users can view a list of completed bookings only
- Access official receipts for completed services
- Review final service details and total payment amount

Receipt Rules:

- Receipts are available only for bookings that have been marked as Completed by the technician

- Once a booking is Completed and Paid, the receipt is locked and cannot be edited

To view transactions:

1. Navigate to Transactions from the user dashboard.
2. Select a completed booking.
3. View or download the receipt associated with the booking.

### Account Management

- Update personal information
- Change account password
- Manage contact details

### Technician Manual

This section guides technicians in managing assigned jobs, updating service details, and confirming service and payment status in real time.

### System Requirements

- Web browser (Chrome, Edge, or Firefox)
- Stable internet connection
- Technician account credentials

### Technician Login

1. Open the technician login page.

2. Enter technician email and password.
3. Click Login to access the technician dashboard.

### Technician Navigation

The technician menu provides access to the following features:

- **Dashboard** – Overview of assigned and ongoing service jobs
- **Repair History** – View completed and past repair services
- **Profile** – Manage technician account information

### Booking Service Management

Technicians are responsible for handling real-time service execution and updates.

Steps:

1. Click the dashboard.
2. Select an assigned booking.
3. Review customer information and reported issues.
4. Select or update the following as needed:
    - Service type
    - Service details
    - Components used
    - Service amount
5. Add additional services if unforeseen issues are discovered during repair.
6. Save changes after updating service details.

## Completing Service and Payment Confirmation

Technicians are solely responsible for updating service and payment status.

Steps:

1. Open the assigned booking.
2. Update the service status to Completed once work is finished.
3. Update the payment status to Paid after receiving payment.

System Behavior:

- Once the technician sets the booking status to Completed and payment status to Paid:
  - The booking is automatically marked as completed and paid
  - Status and payment dropdown options become disabled for that booking
  - No further edits are allowed by the technician or administrator

## Administration Manua

This guide will help administrators navigate and utilize all features of the admin system for monitoring bookings, assigning technicians, and managing records.

### System Requirements

- Desktop or laptop computer
- Web browser (Google Chrome or Microsoft Edge)
- Stable internet connection
- Administrator account credentials

### Accessing the Admin Panel

1. Navigate to the admin login page.
2. Enter admin username and password.
3. Click Login to access the admin dashboard.

### Getting Started

After logging in, the admin dashboard will display an overview of system activity and statistics.

### Navigation

The sidebar menu provides access to the following features:

- **Dashboard** – System overview and statistics
- **Manage Bookings** – View bookings, assign technicians, and monitor booking status
- **Technicians** – Manage technician accounts and availability
- **Manage Transactions** – Review payments and confirmed transactions

- **Sales Report** – Generate and view sales and revenue reports

### Dashboard Overview

The admin dashboard provides a quick summary of daily system activity:

- **Estimated Total Sales (Today)** – Displays the estimated total sales for the current day only
- **Total Bookings (Today)** – Shows the total number of bookings created for the current day

Note:

- The dashboard only displays daily summaries
- Detailed and cumulative sales and booking data are available in the Sales Report section

### Booking Management

Administrators oversee all service requests but cannot directly update booking status or payment status during real-time service execution.

Administrator responsibilities include:

- Reviewing submitted bookings
- Assigning technicians manually when needed
- Adjusting service schedule if required
- Confirming additional services added by technicians

***Technician Assignment and Schedule Management***

Steps to assign a technician and service schedule:

1. Select a booking from the booking list.
2. Click Assign Technician.
3. Choose an available technician based on specialization.
4. Set or update the service date and time.
5. Click Save Assignment

***Additional Service Confirmation and Receipt Update***

- When a technician adds extra services during real-time repair, the admin will confirm the additional services
- Once confirmed, the system generates an updated receipt reflecting the added service and revised total

***Receipt Locking Rule***

- Receipt generation for a booking becomes locked once the technician confirms:
    - Booking status as Completed
    - Payment status as Paid
- No further modifications are allowed after confirmation

## Sales Report

The Sales Report section provides detailed and cumulative sales and booking information.

Steps to Generate Sales Report:

1. Navigate to Sales Report from the admin menu.
2. Select the desired date range.
3. Click Generate Report
4. Review total sales and total bookings displayed in the report

**Account and System Management**

- Manage user and technician accounts
- Monitor system activity
- Ensure data accuracy and security

**Appendix G:** Similarity Index (Turnitin)

**Appendix H: Grammarian Certificate**

GRAMMARIAN CERTIFICATE

This is to certify that the undersigned has reviewed and gone through all the pages of the proposed capstone project entitled **"SmartLink: A Unified Platform for IT Services and Customer Support"**

prepared by:

Buhanghang, Jay

Diamante, James Michael H.

Madera, Adrian

Quiñones, Michael John S.

Yu, John Wilson

against the set of structural rules that govern the composition of sentences, phrases, and words in the English language.

Verified and Certified by:

Marissa Quiñones Puli

Grammarian/English Teacher

PRC License No.: 2085441

Date signed: July 16, 2026 LPT. MAEd

**Appendix I: Curriculum Vitae**

# James Michael Diamante

IT Undergraduate

I strive to embody professionalism and empathy, Motivated and driven undergraduate student seeking an internship to gain practical experience and apply academic knowledge in a professional environment. I am a quick learner with strong communication skills, a positive attitude, and a strong determination to grow.

## Contact


**Phone**
0949 123 2783

**Email**
jmdiamante20@gmail.com

**Address**
3061A Cordillera St. Sta Mesa, Manila


## Education

2022 - Present
**Bachelor of Science in Information Technology**
National Teacher's College

2020 - 2022
**Humanities and Social Sciences**
Major Manuel A. Aaron Memorial National High School

2016 - 2020
**Junior High School**
Pasay City East High School

## Expertise

- Communication skills
- Team Collaboration
- Time management
- Problem-solving

## Skills

## Technical Skills

**Software & Tools**

- Basic MS Word, Powerpoint, Excel
- Canva
- Basic knowledge in Figma
- Visual Studio Code
- XAMPP

**Programming**

- Basic PHP
- Basic Javascript
- Python
- Basic HTML & CSS
- Basic Framework: Bootstrap/ Tailwind CSS
- Basic MySQL Database Management

## COURSE CERTIFICATION

**Coursera**

- Introduction to Software Engineering
- Developing Front-End Apps with React
- Developing Back-End Apps with Node.js and Express
- Embedded Systems Software and Development Environments
- Cyber-Physical Systems: Modeling and Simulation
- Simulation Models for Decision Making
- Responsive Web Design

# ADRIAN MADERA

IT UNDERGRADUATE

## EDUCATION

2022 - Present
**BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**
NATIONAL TEACHERS COLLEGE

2017 - 2019
**ABM - SENIOR HIGH SCHOOL**
ARELLANO UNIVERSITY

## SOFT SKILLS

- Communication Skills
- Adaptability
- Problem Solving
- Time Management
- Critical Thinking

## CONTACT


09552856243

adrianmadera456@gmail.com

Damka St. Old Sta. Mesa Manila


## PROFILE INFO

A future Information Technology professional with a strong academic foundation and a keen interest in emerging technologies. Skilled at quickly mastering new systems and applying adaptable solutions to enhance efficiency and drive impactful results.

## SKILLS

### TECHNICAL SKILLS

**SOFTWARE AND TOOLS**

-Microsoft Office (Word, Powerpoint, Excel)
-Google Workspace (Docs, Sheet)
-Figma UI/UX)
-Visual Studio Code (VSCode)

**PROGRAMMING AND DEVELOPMENT**

-Basic PHP
-Basic JAVA
-Python
-Basic Knowledge in Web Developing

## CERTIFICATION

**COURSERA**

Simulation Models for Decision Making
Developing Back-End Apps with Node.js and Express
Introduction to Predictive Modeling
Blockchain Basics
Developing Front-End Apps with React
Modelling and simulation of mechanical systems

# JOHN WILSON YU

IT Undergrad

## About Me

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

## Contact

951-399-0093

422000361@ntc.edu.ph

744 Camba, Binondo Manila

## Skills

- **Software**
- MS Word
- MS Excel
- Software installation & updates
- **Hardware**
- Peripheral setup
- System maintenance
- BIOS & system settings

## Language

- English
- French

## Education

**2018-2019**

**MACEDA INTEGRATED SCHOOL**

Senior High School

**2022 - Present**

**NATIONAL TEACHERS COLLEGE**

Bachelor of Science Information Technology

## Professional Experience

**(2020 -2023)**

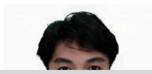
**COMPUTER HARDWARE & SOFTWARE TECHNICIAN**

- Installed, configured, and maintained computer hardware including desktops, laptops, printers, and peripherals
- Diagnosed and resolved hardware issues (RAM, HDD/SSD, motherboard, power supply)
- Installed and updated operating systems (Windows) and essential software

**2020 - 2023**

**COMPUTER OPERATOR**

- Managed data entry, data cleaning, and data validation
- Generated charts, reports, and summaries for analysis
- Maintained accurate records and ensured data integrity



**JAY BUHANGHANG**

# Michael John Quinones

IT Support

## About Me

Motivated and detail oriented. Information Technology . Graduate . Capable of working independently as a part of the team in a fast paced environment.

## Contact

09918513176

qmikeljohn0002@gmail .com

55 San Isidro St. Q.C.

## Skills

- Web Design
- Branding
- Graphic Design
- Marketing

## Language

- English

## Education

- **(2022-2026)**
  **NATIONAL TEACHERS COLLEGE**
  Bachelor of Information Technology
  3.65

## Certification

- NC II Computer Systems Servicing
- FreeCodeCamp Certifications
- OJT Training Certificates
- IT-related Certificates